



BobSim

Suspension Kinematics and Compliance (KnC) Characterization

Heave and Roll Sweeps

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– These results include heave and roll sweeps to evaluate key suspension parameters such as camber, toe, caster, KPI, trail, scrub, motion ratio, and jacking characteristics.

KnC Metrics Summary

Heave Gains

Camber	0.4557	rad/m
Toe	-0.0019	rad/m
Caster	-0.0390	rad/m
KPI	0.4551	rad/m
Trail	-0.0011	m/m
Scrub	0.0018	m/m

Anti & Balance

Anti-Dive	2.5	%
Anti-Squat	-5.3	%
Front Anti-Roll	-0.8	%
Rear Anti-Roll	-0.9	%
Anti Balance (F/R)	0.77	-
LLTD (Front)	52.2	%

Roll Gains

Camber	0.7245	rad/rad
Toe	0.0013	rad/rad
Caster	0.0344	rad/rad
KPI	0.7249	rad/rad
Trail	-0.0015	m/rad
Scrub	0.0026	m/rad

Motion Ratios

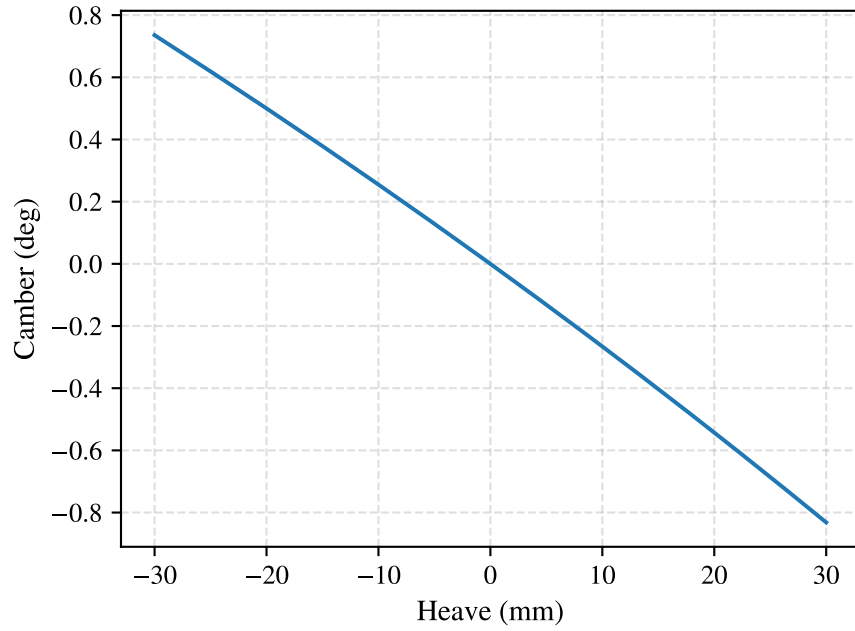
Front MR	0.991	-
Rear MR	1.251	-
Front Bar MR	0.127	-
Rear Bar MR	0.155	-

Roll Stiffness

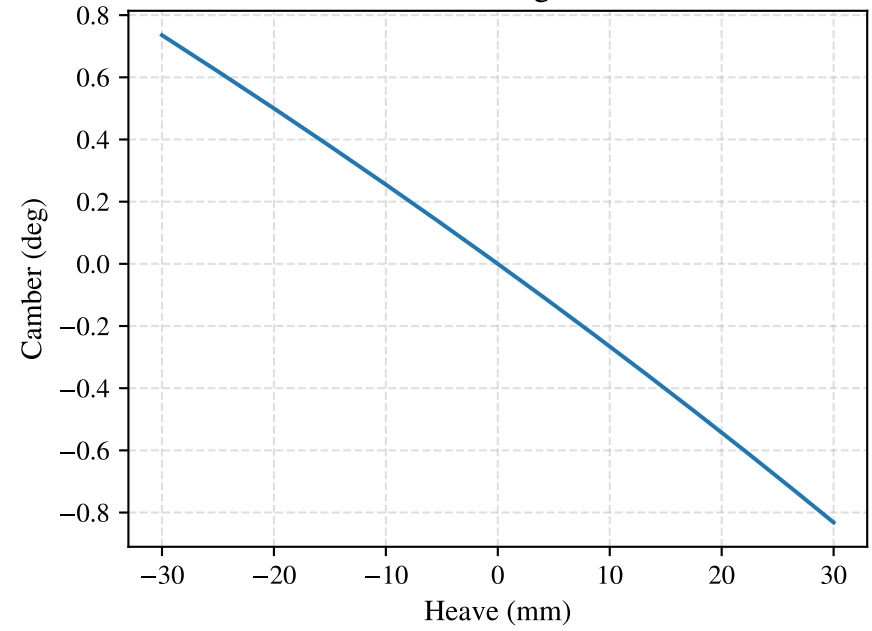
Spring Front	21983	Nm/rad
Spring Rear	12888	Nm/rad
ARB Front	46328	Nm/rad
ARB Rear	49753	Nm/rad
Total Front	68311	Nm/rad
Total Rear	62641	Nm/rad

Camber vs Heave

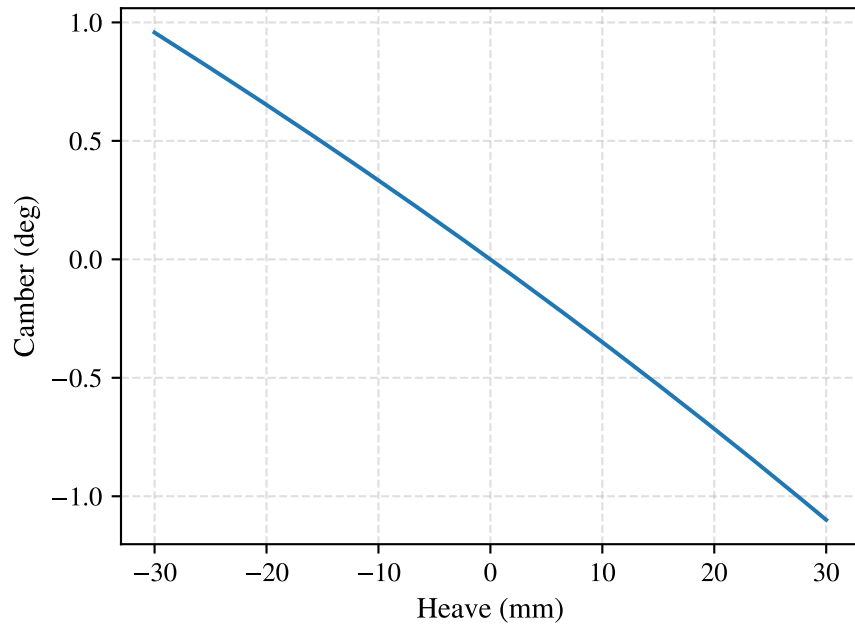
Front Left



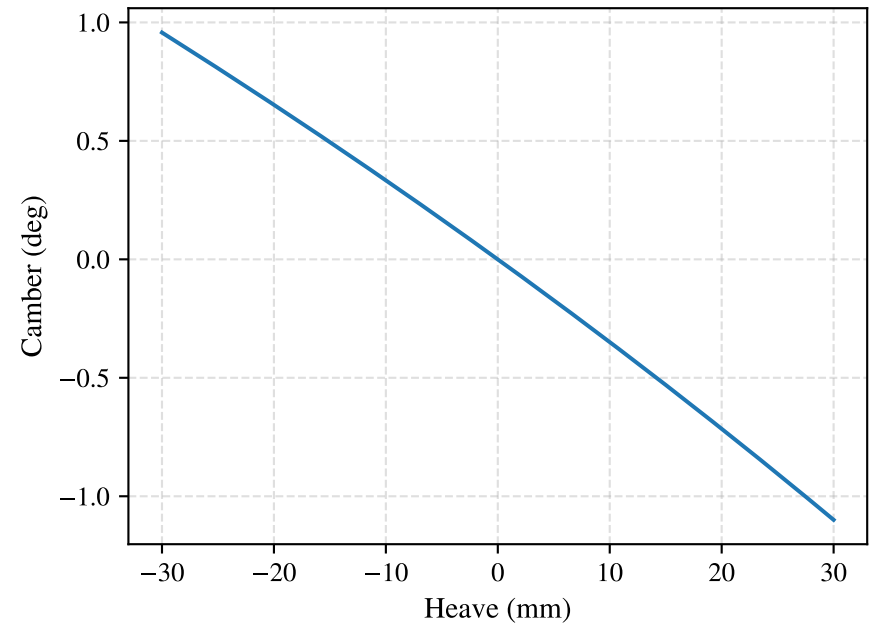
Front Right



Rear Left

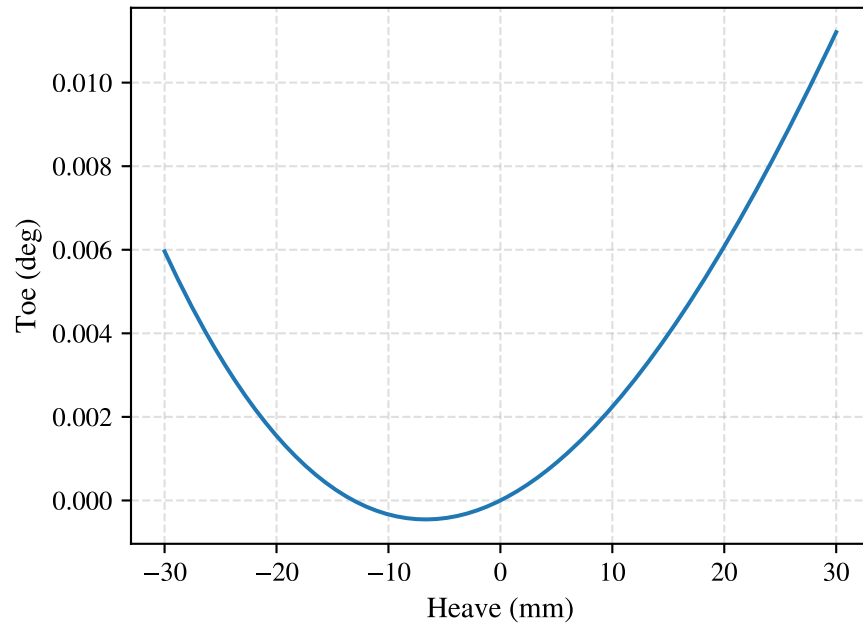


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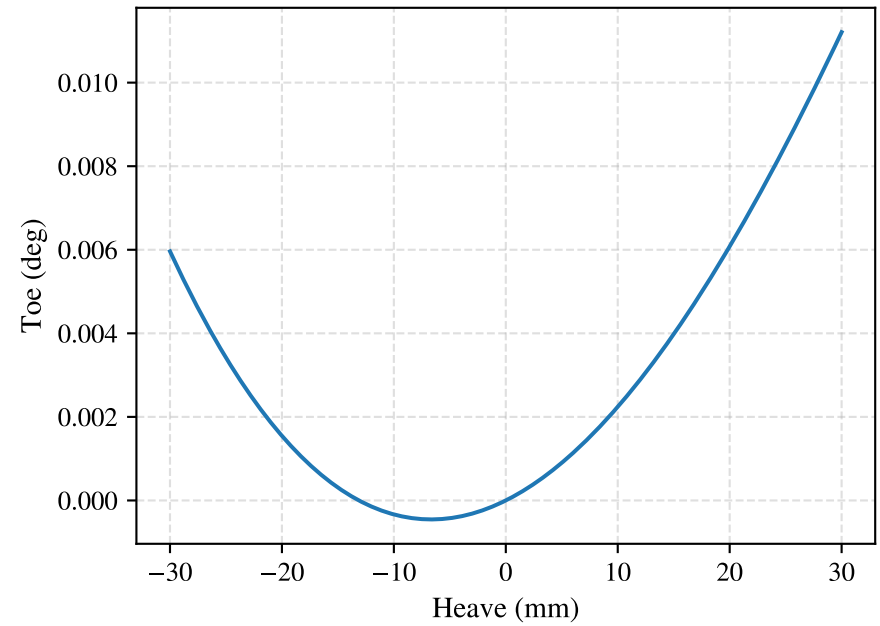


Toe vs Heave

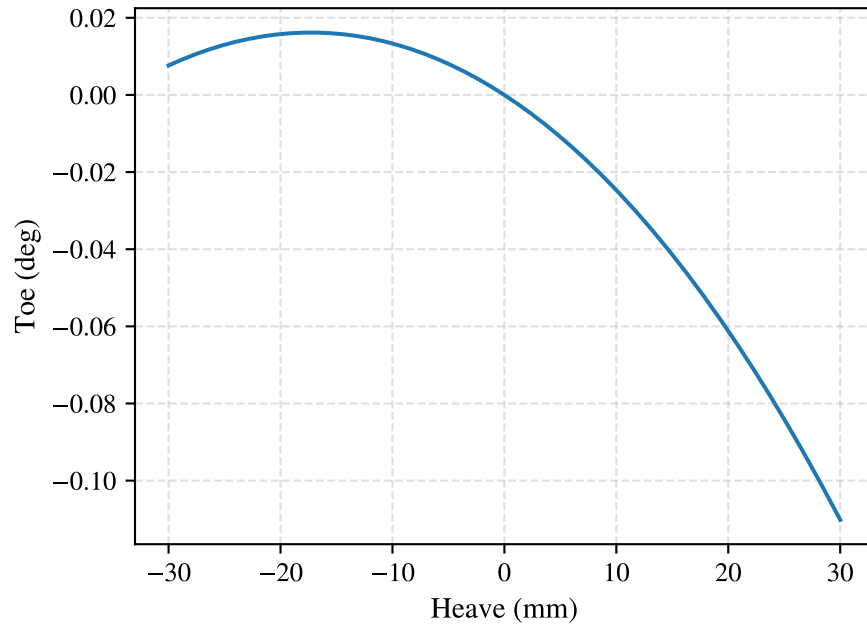
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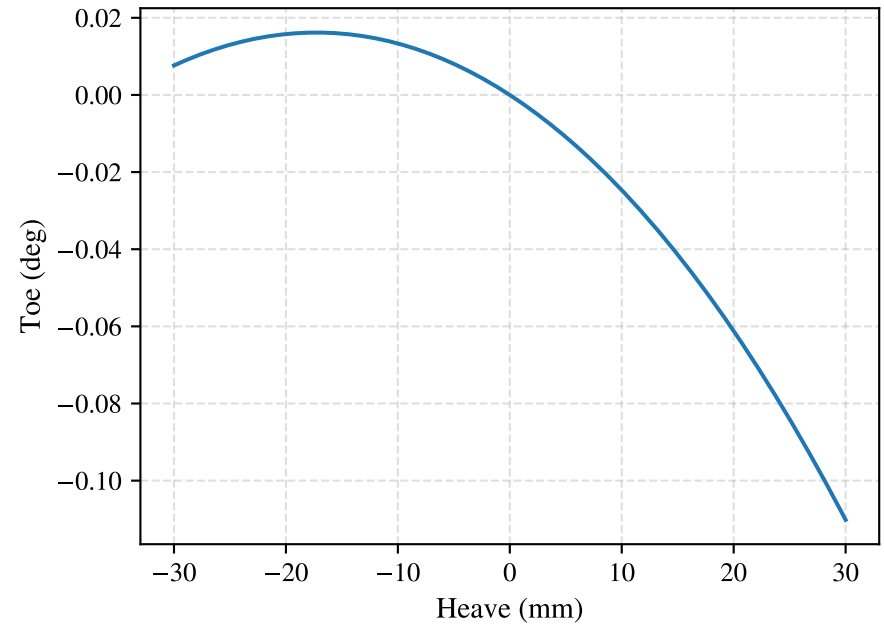
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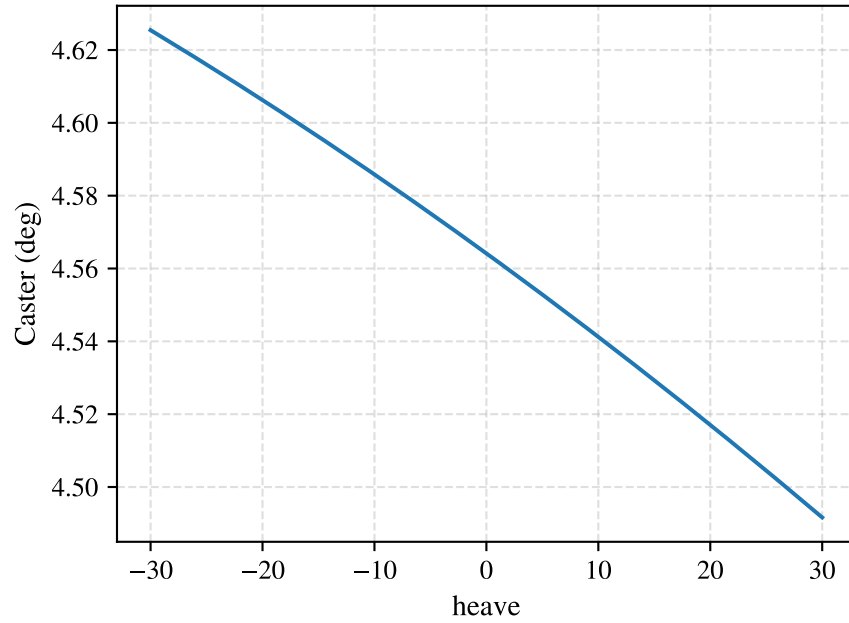


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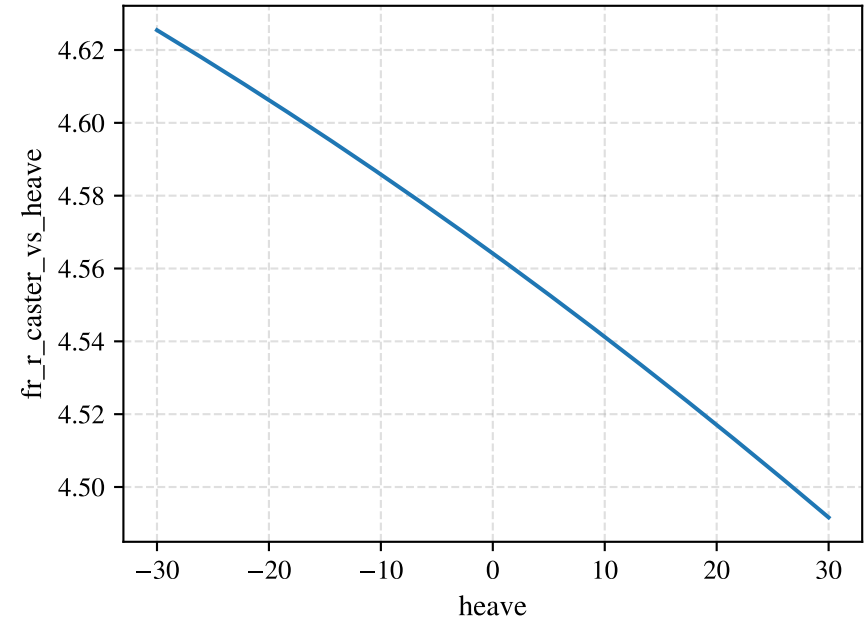


Caster vs Heave

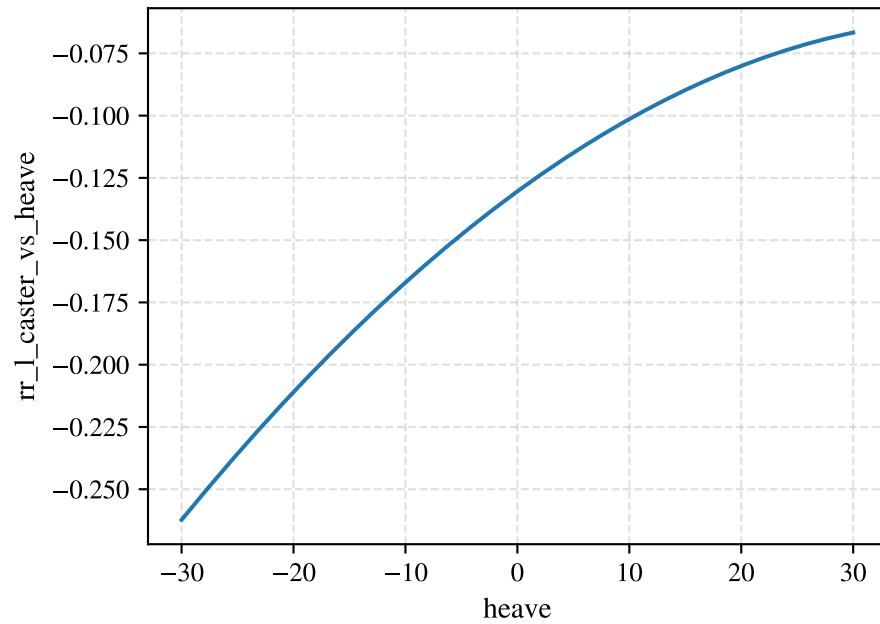
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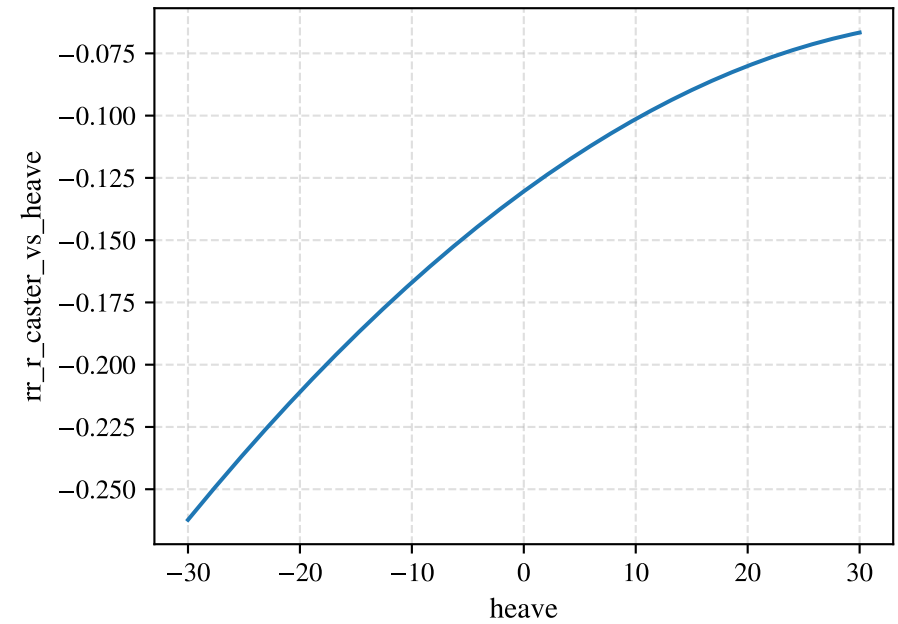
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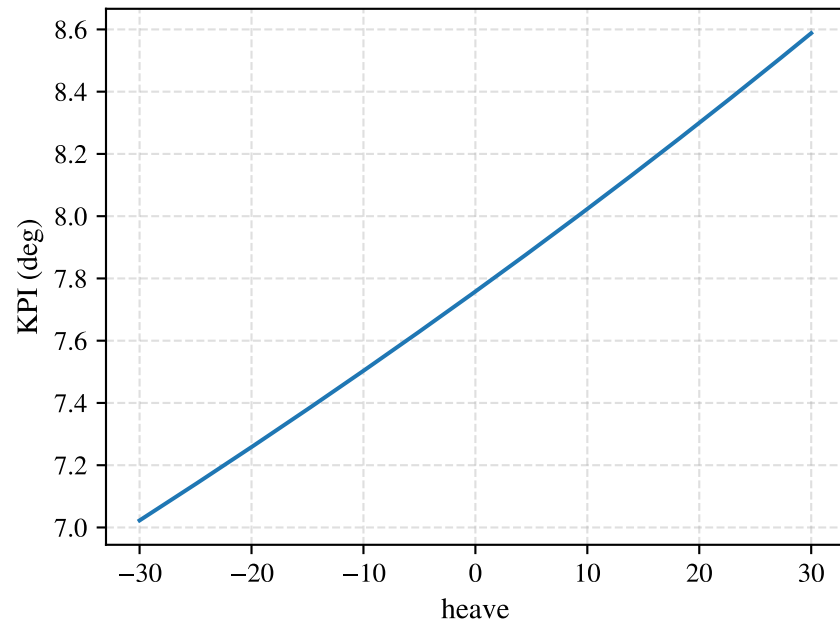


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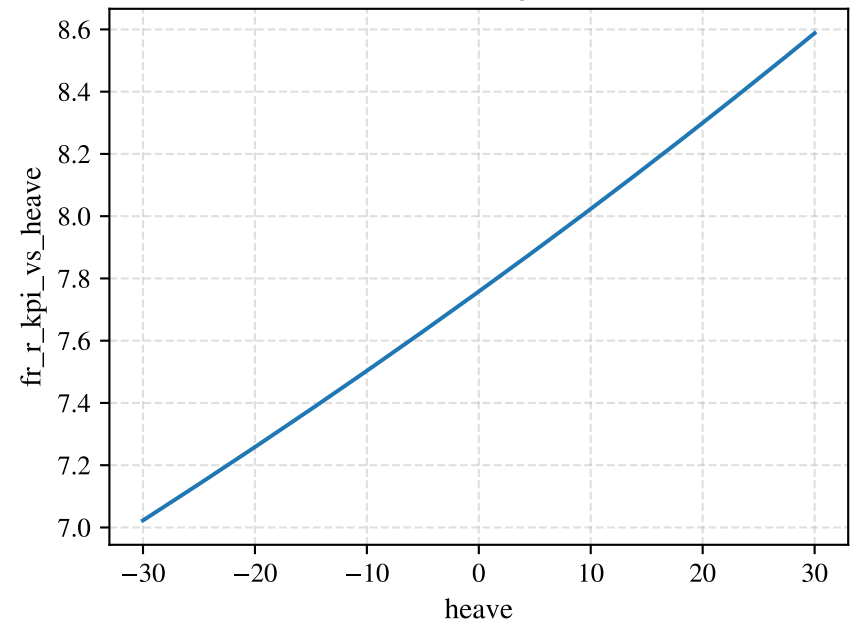


KPI vs Heave

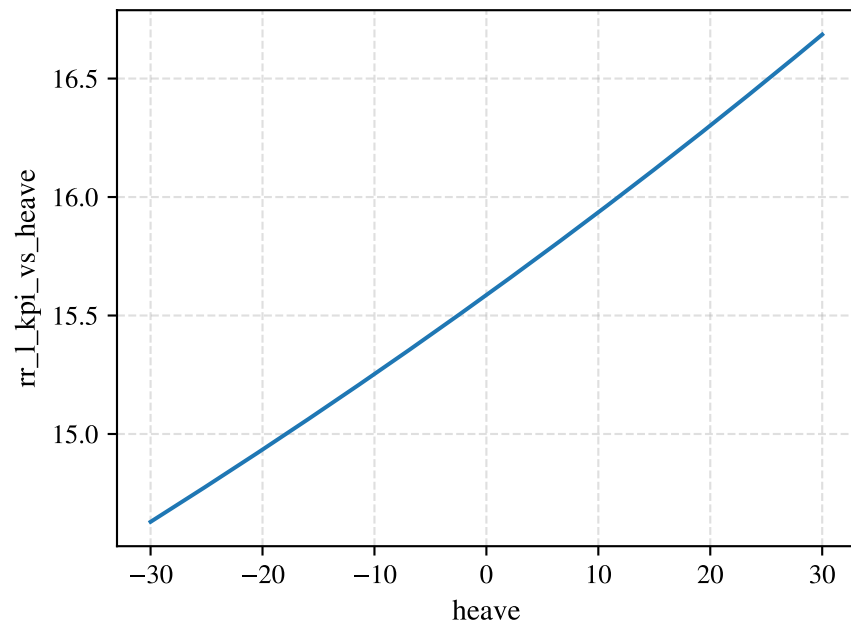
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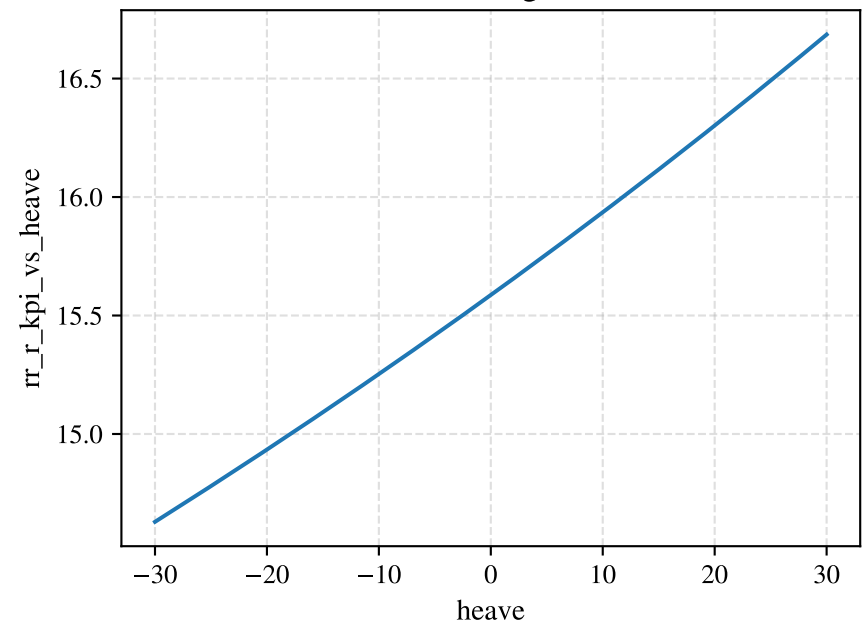
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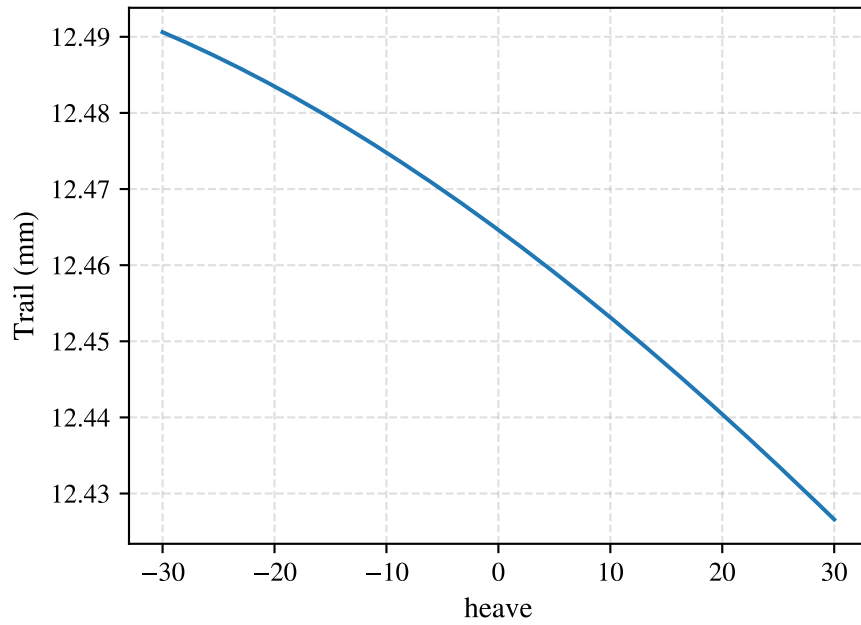


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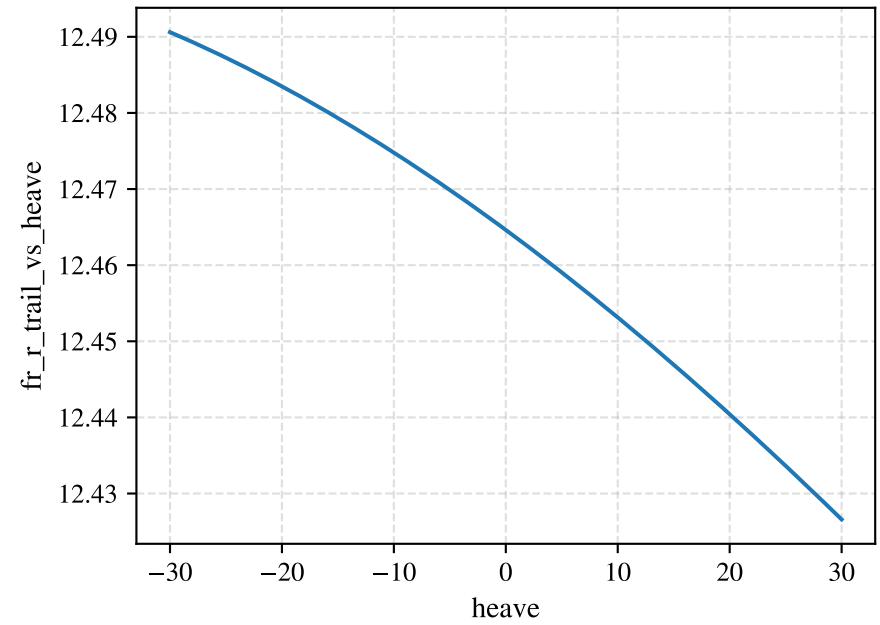


Mechanical Trail vs Heave

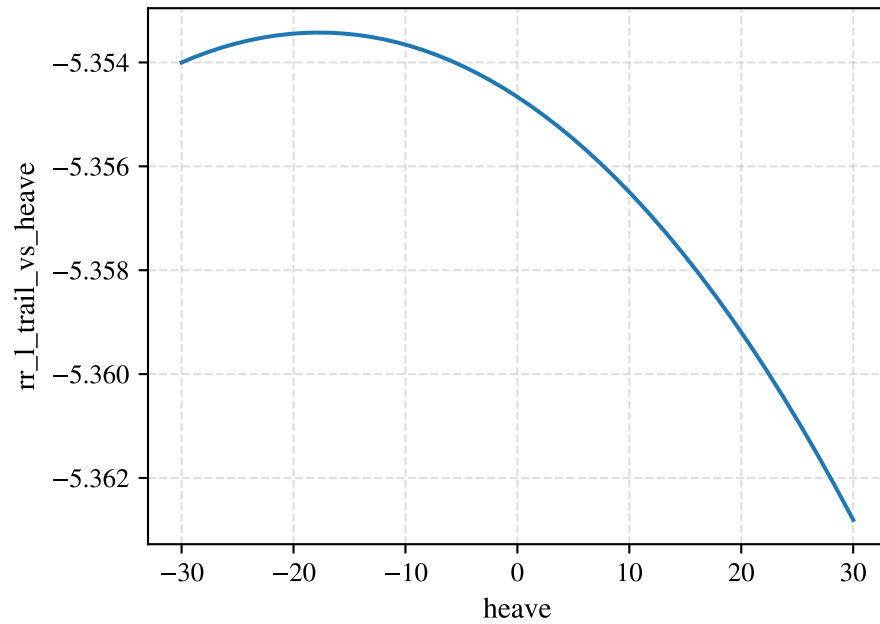
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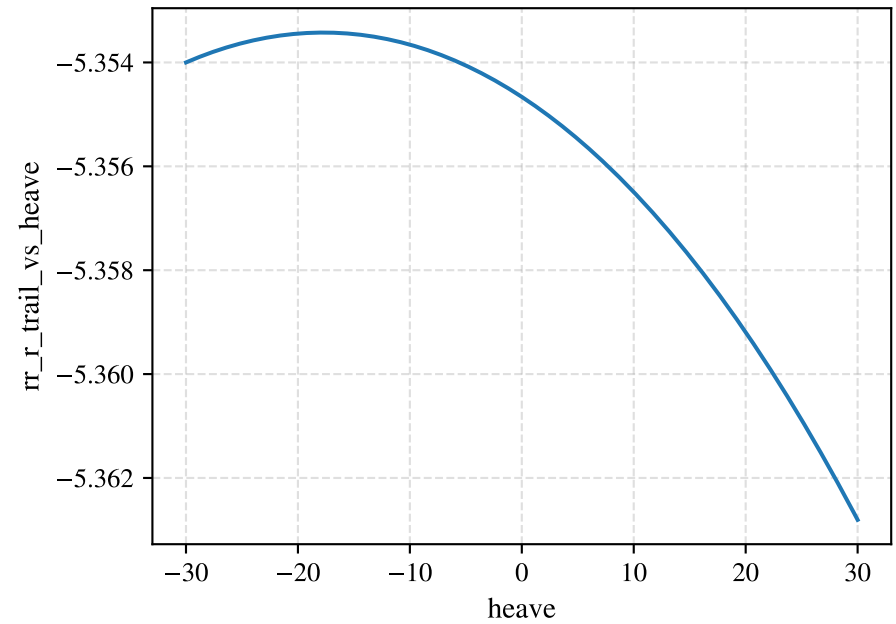
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Rear Left

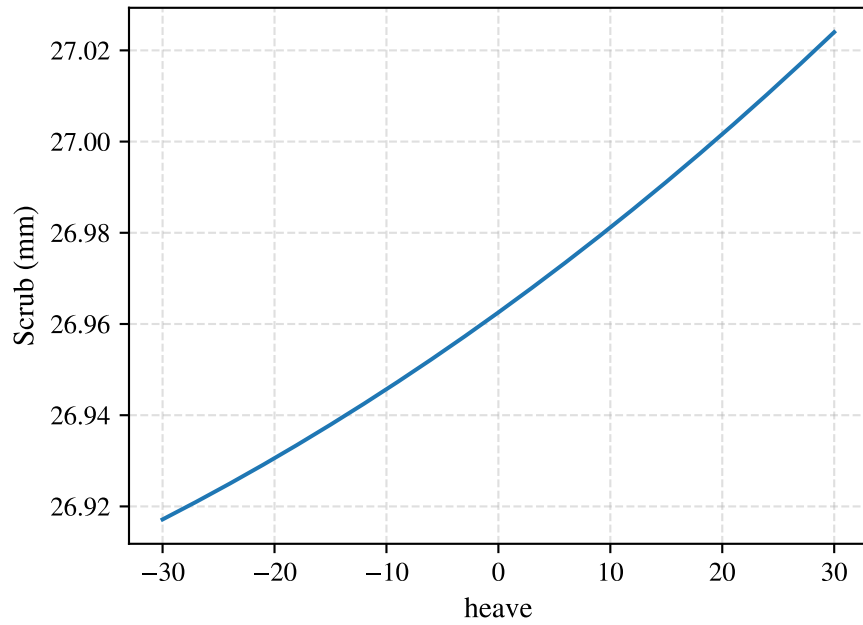


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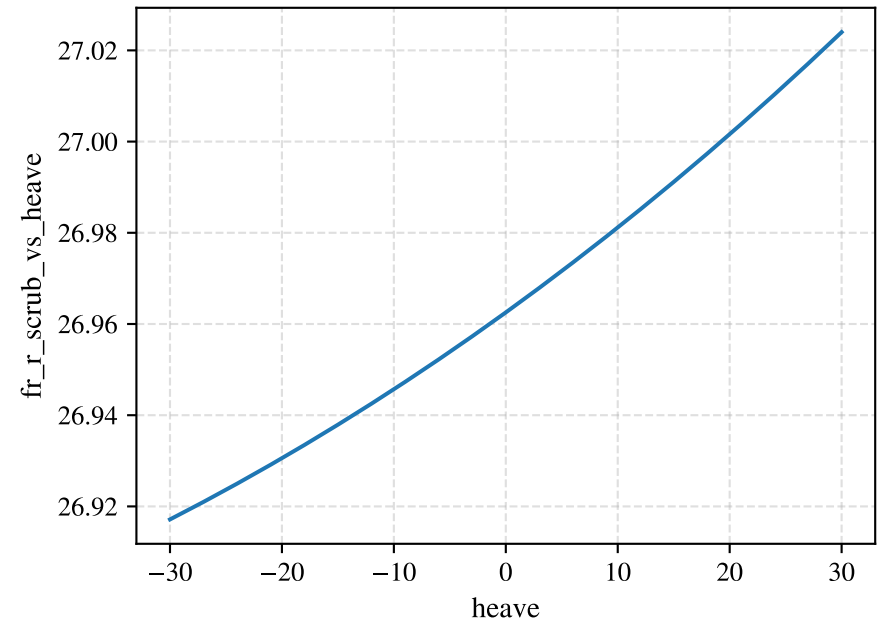


Mechanical Scrub vs Heave

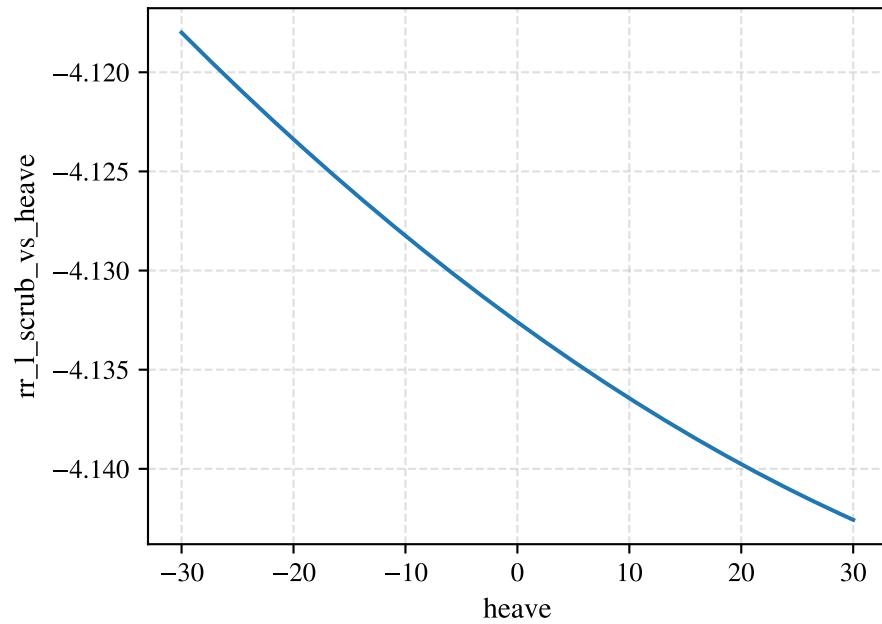
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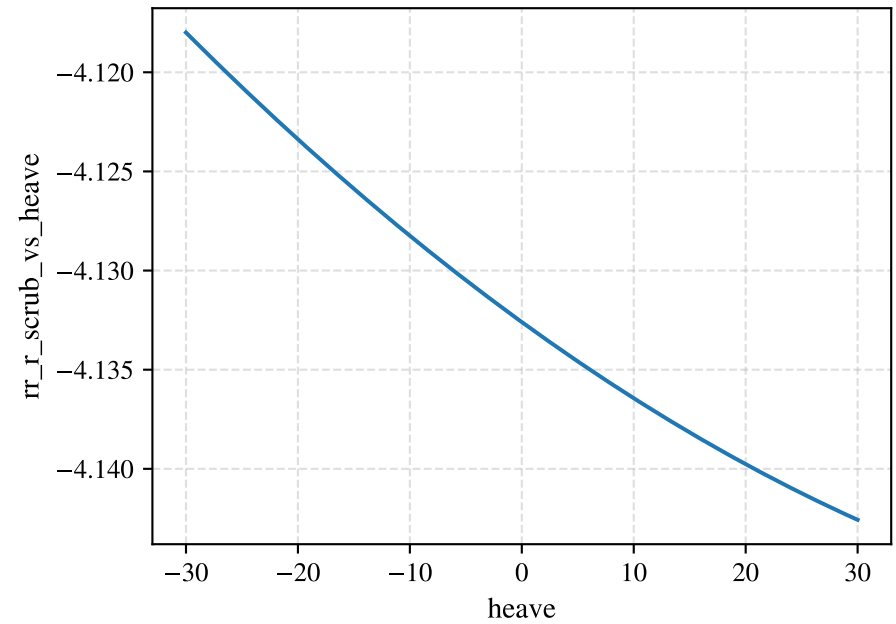
Front Right



Rear Left

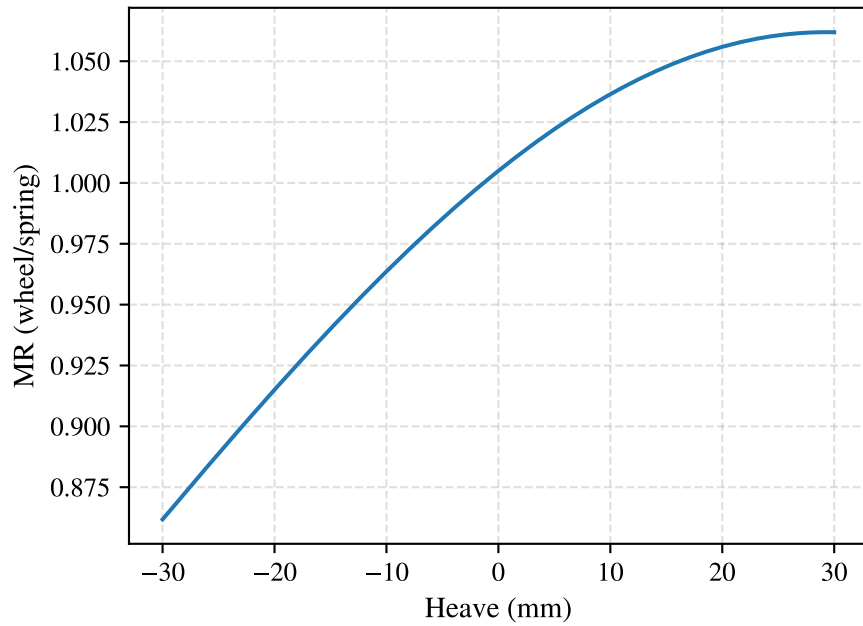


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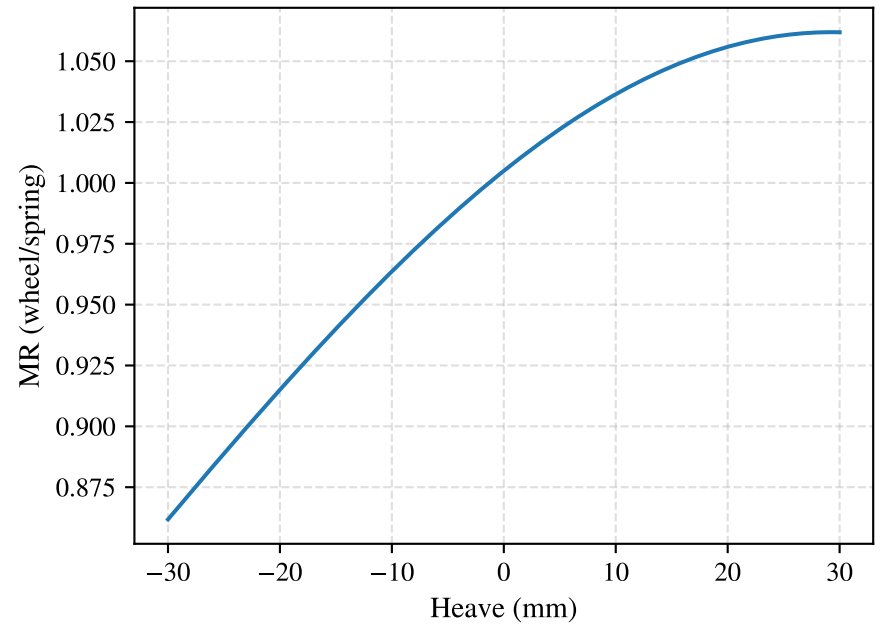


Motion Ratio vs Heave

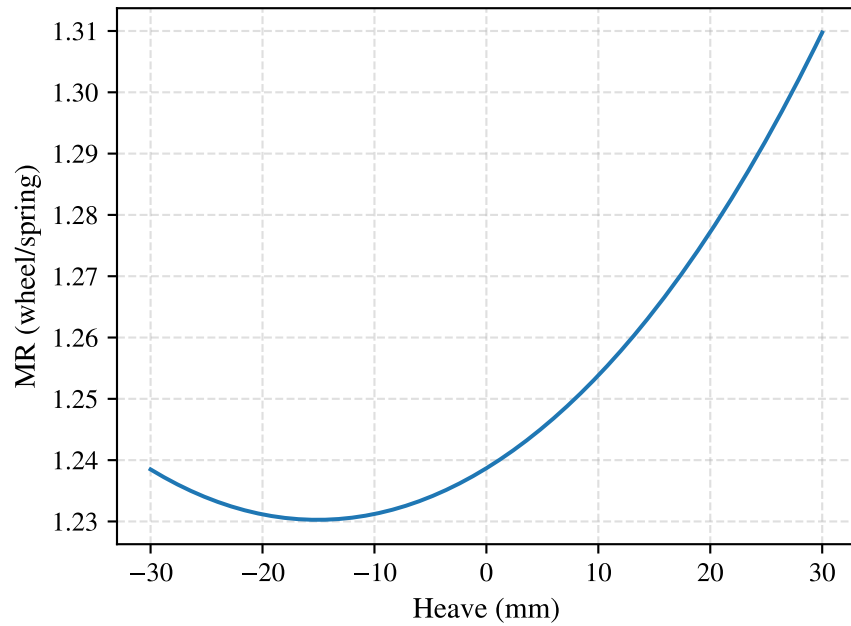
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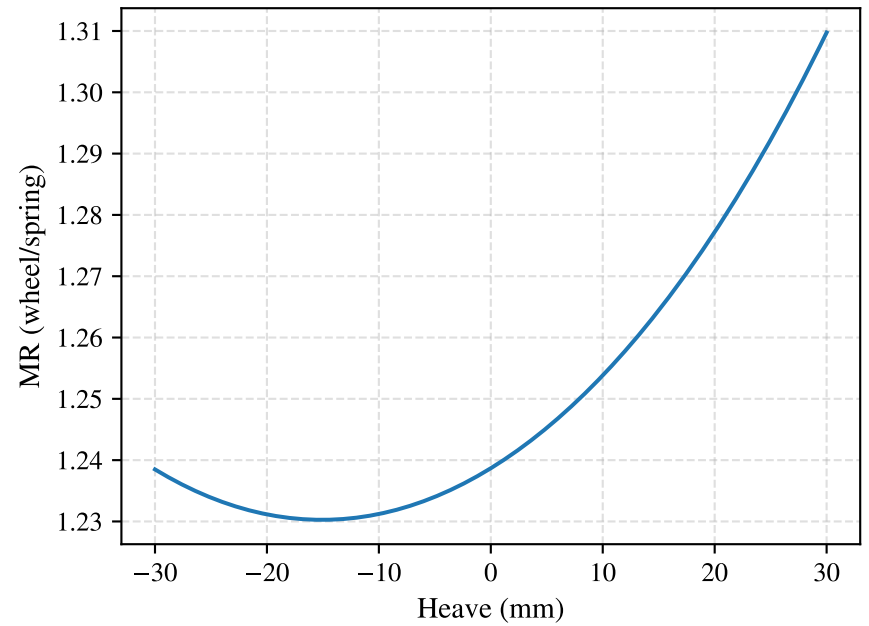
Front Right



Rear Left

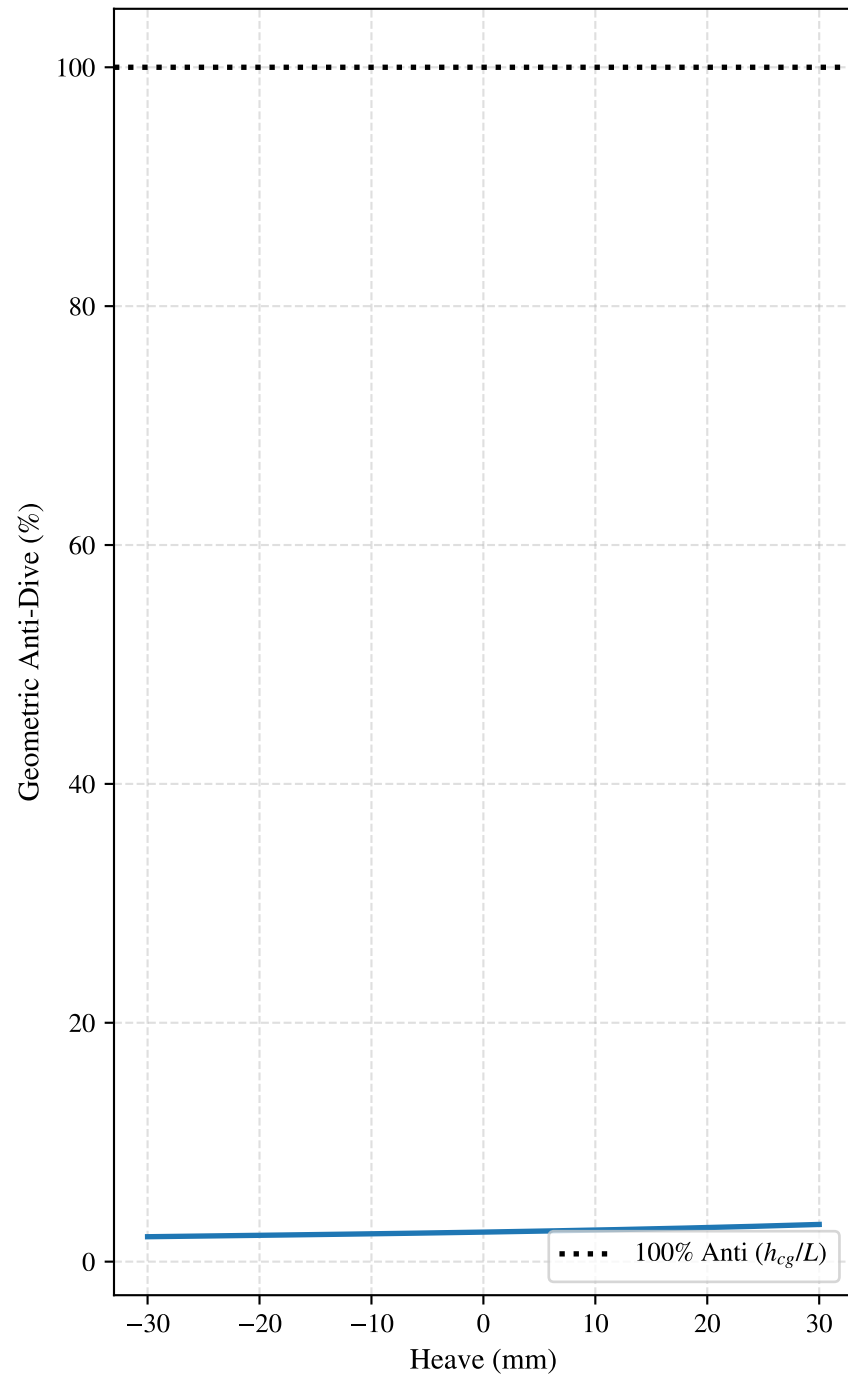


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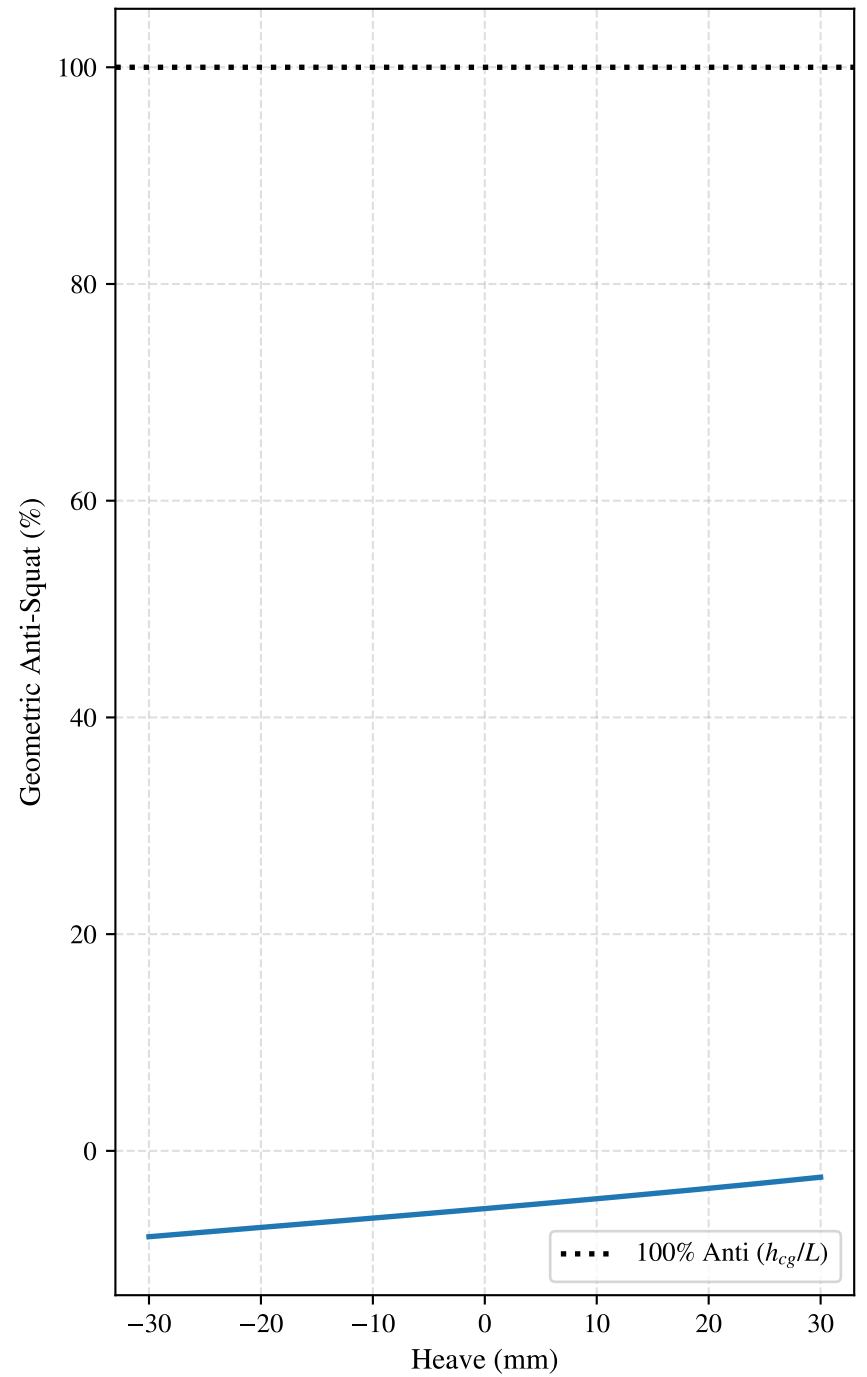


Anti vs Heave

Front

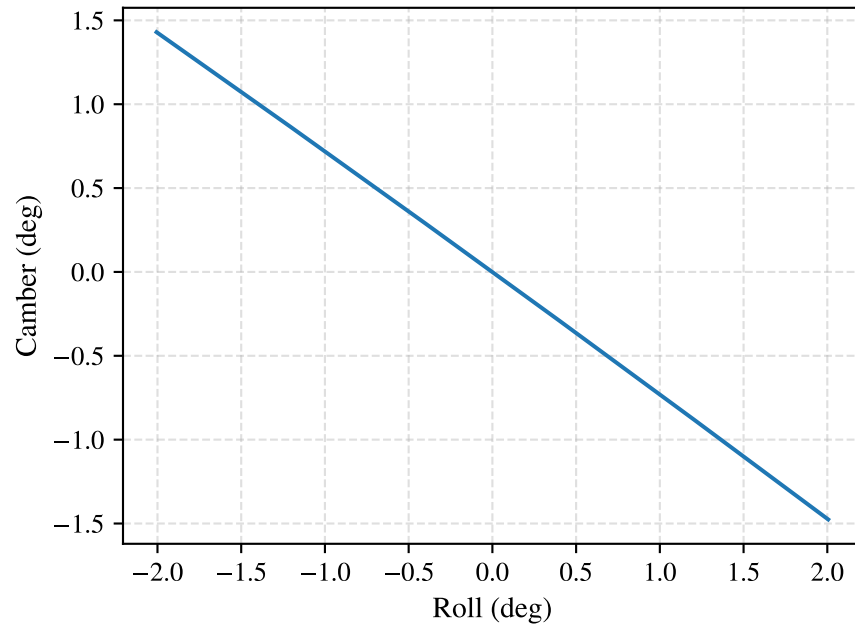


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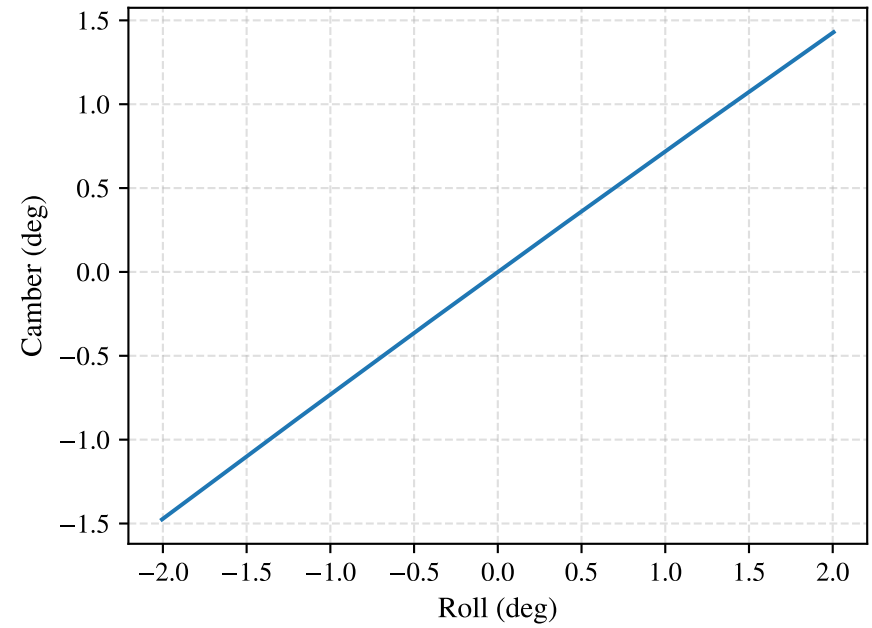


Camber vs Roll

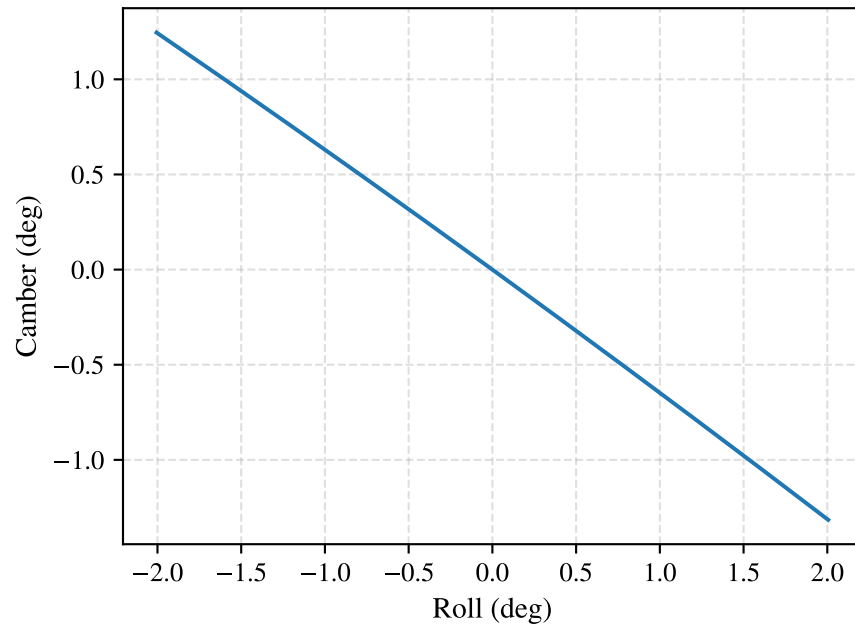
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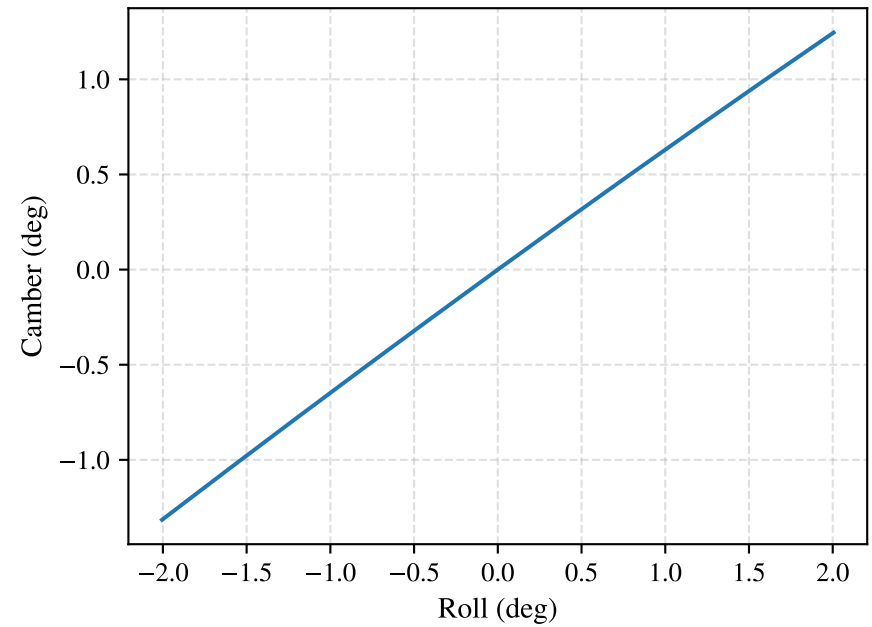
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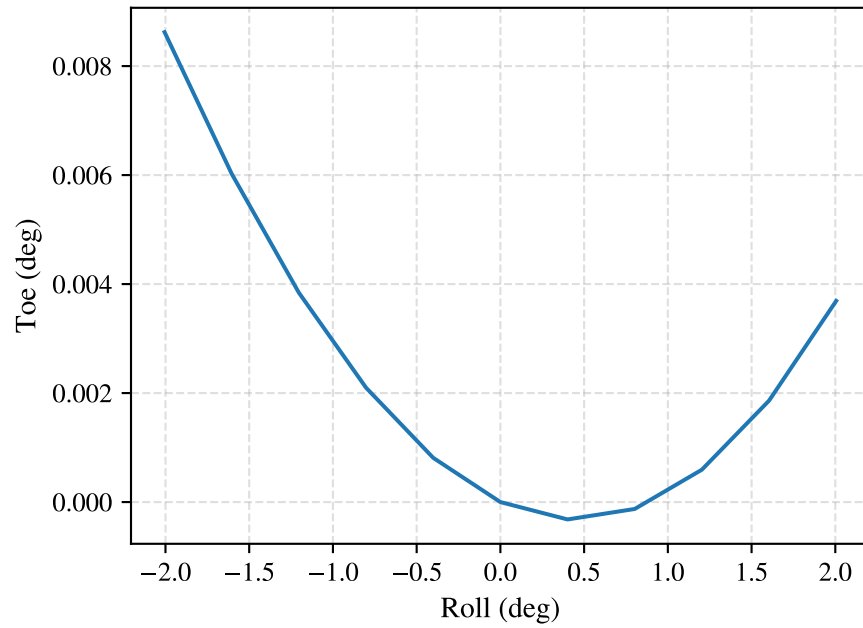


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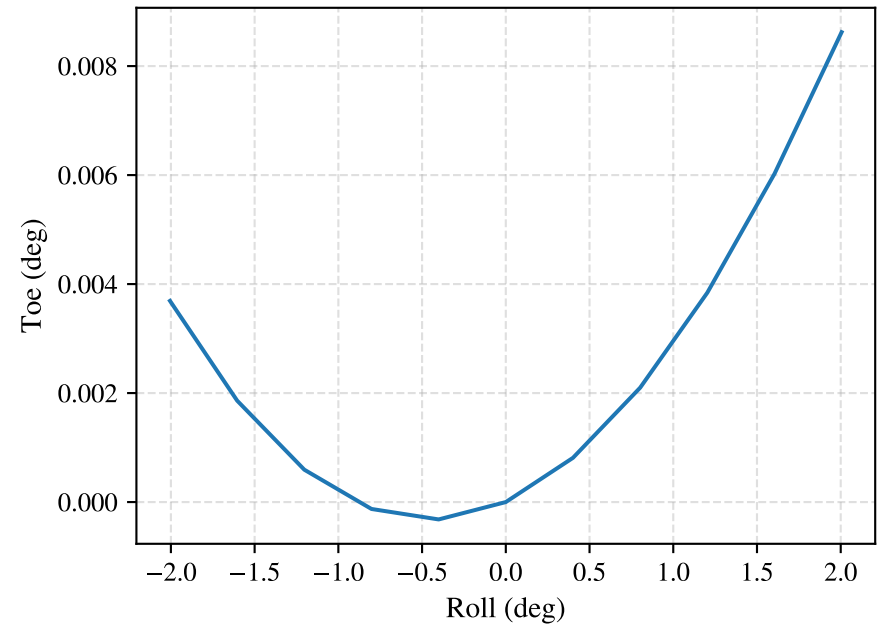


Toe vs Roll

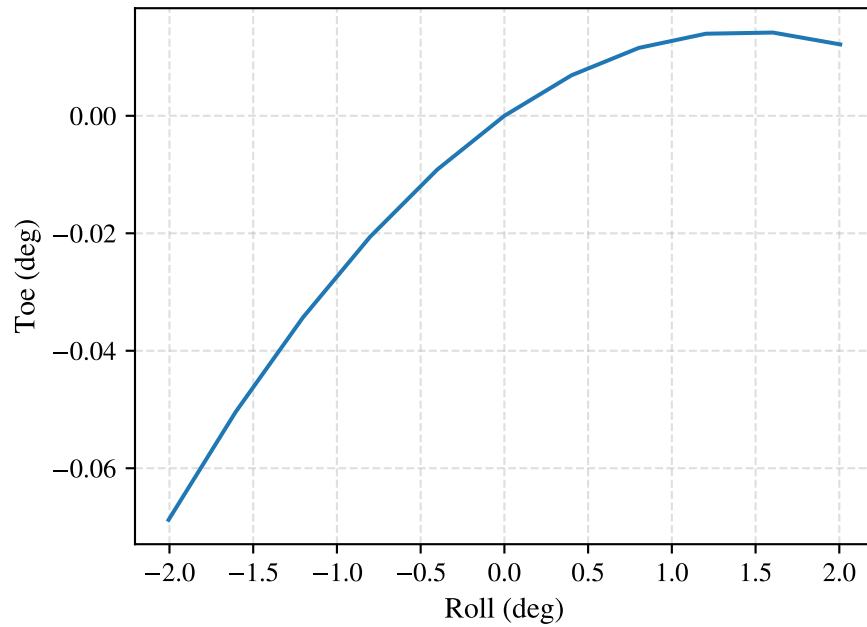
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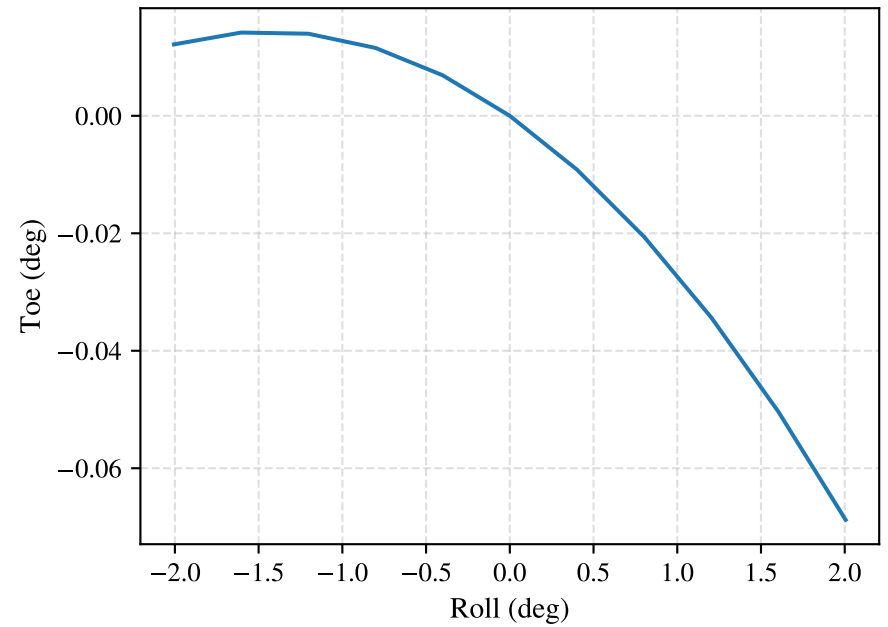
Front Right



Rear Left

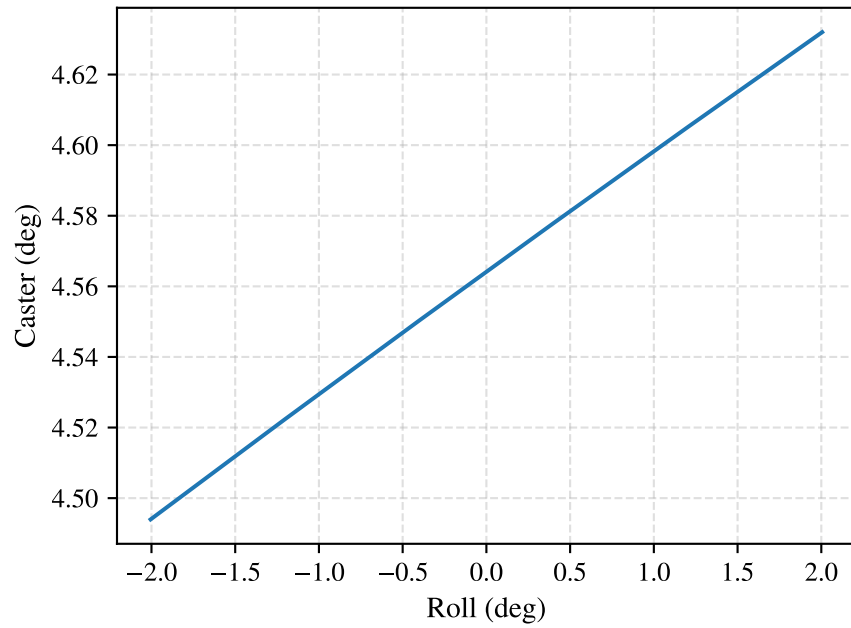


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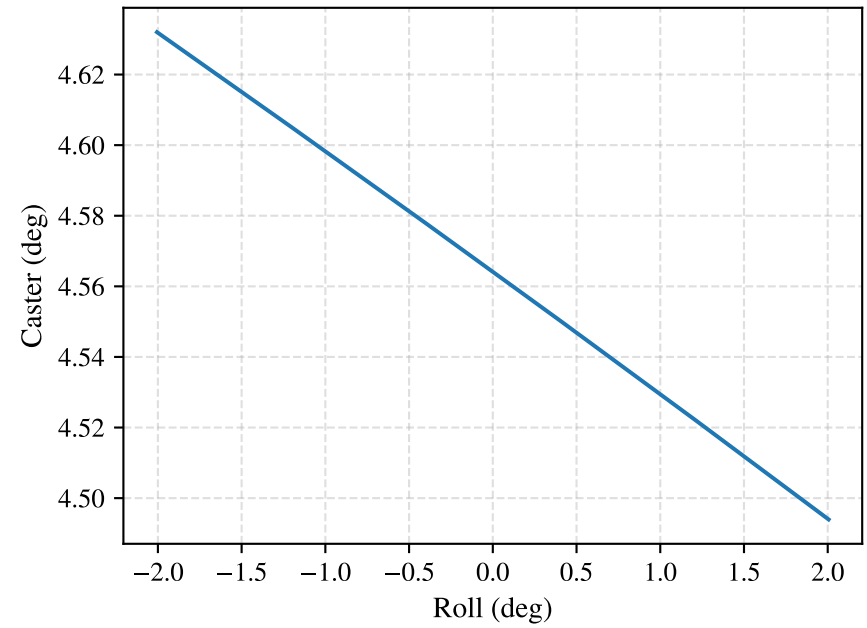


Caster vs Roll

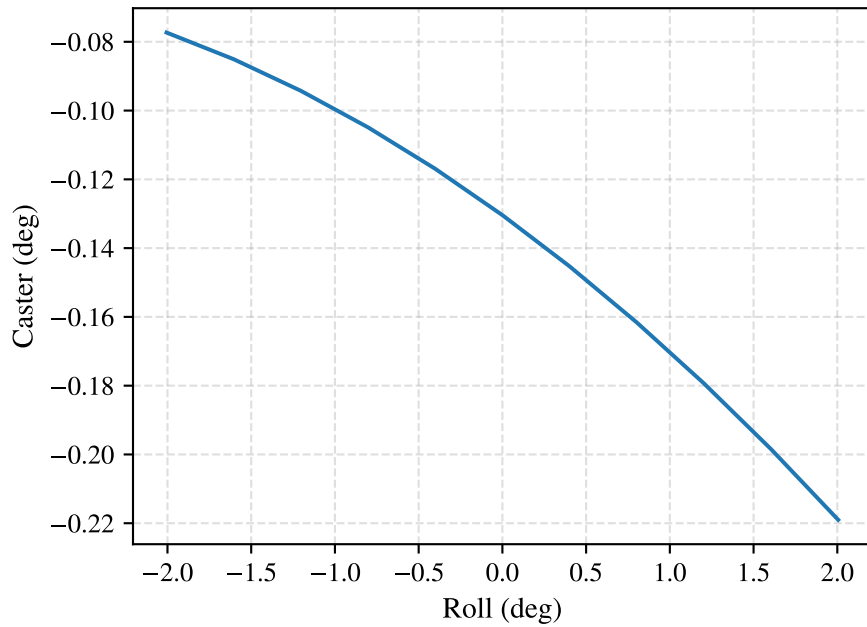
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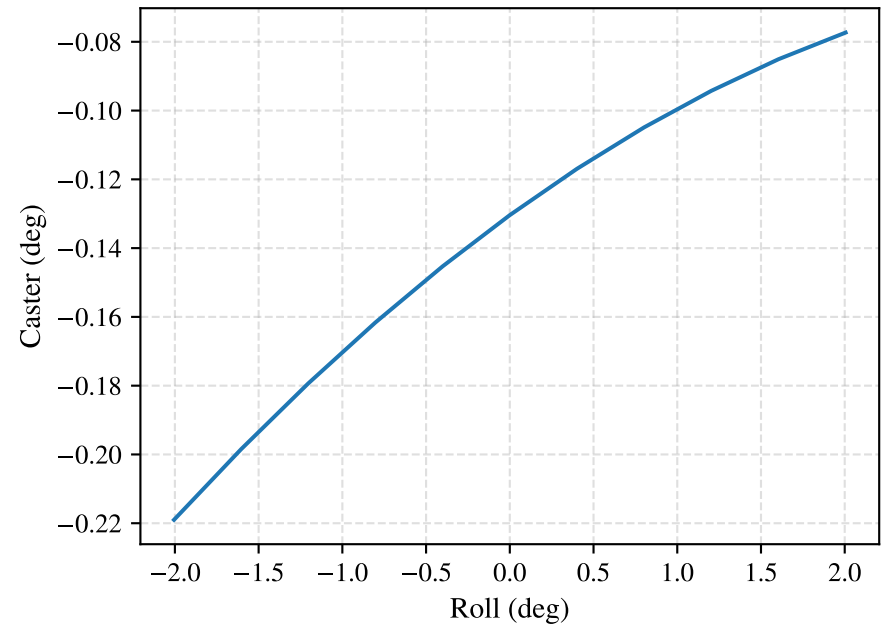
Front Right



Rear Left

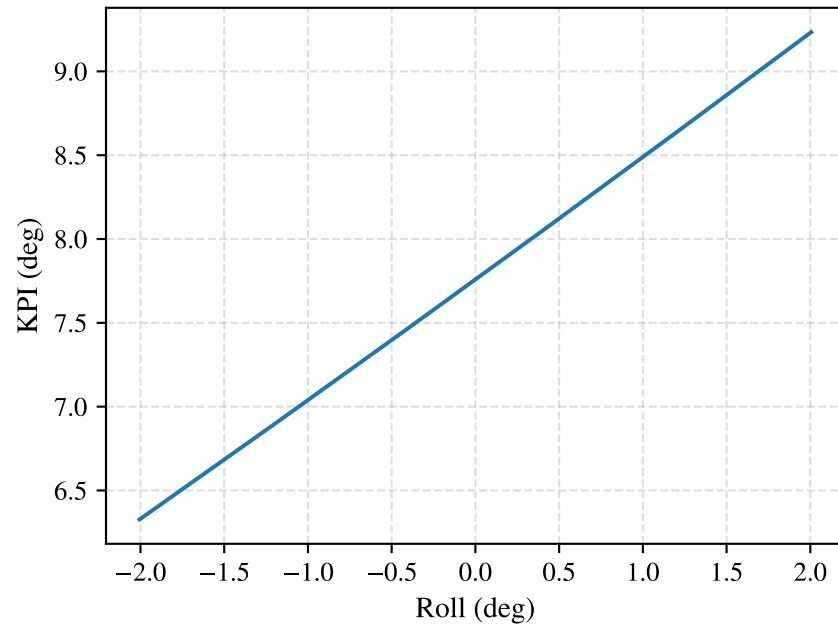


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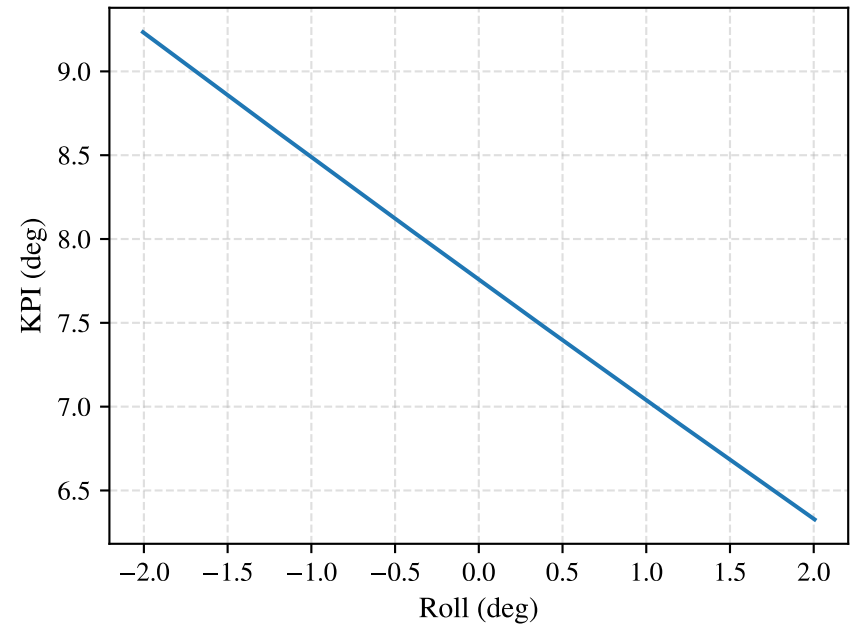


KPI vs Roll

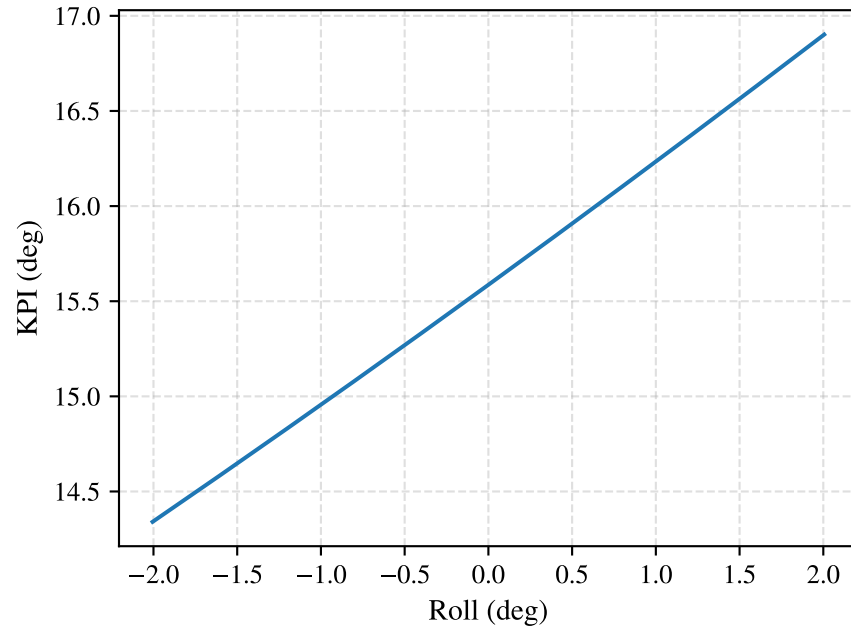
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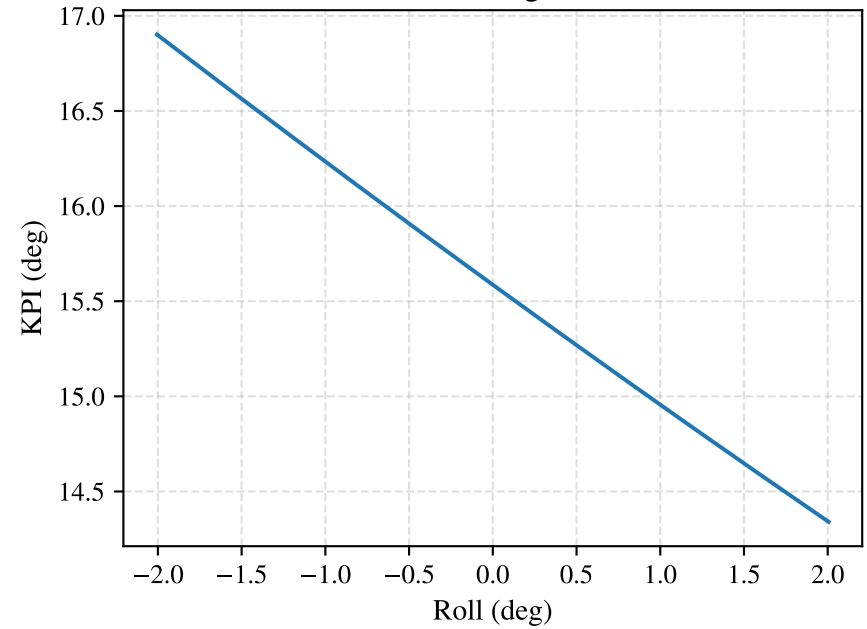
Front Right



Rear Left

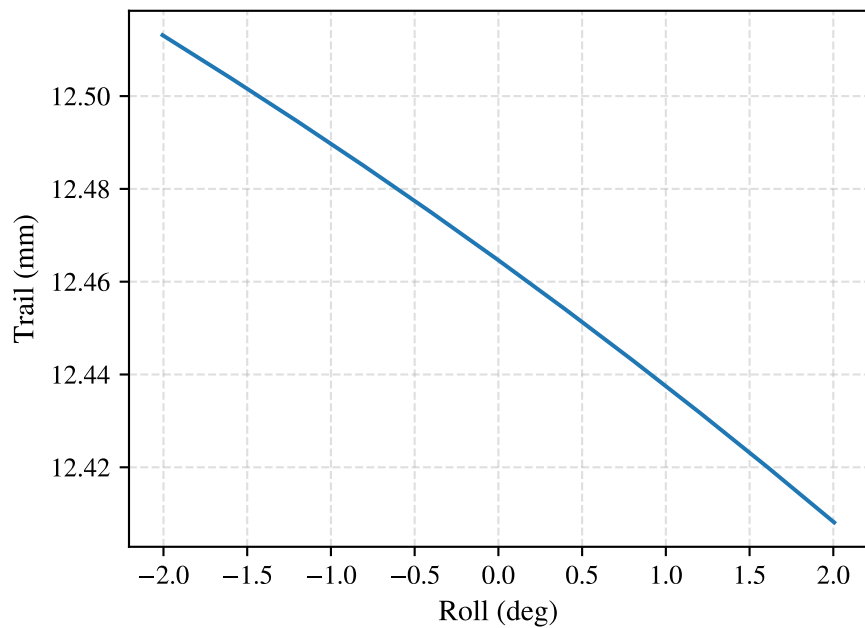


Rear Right

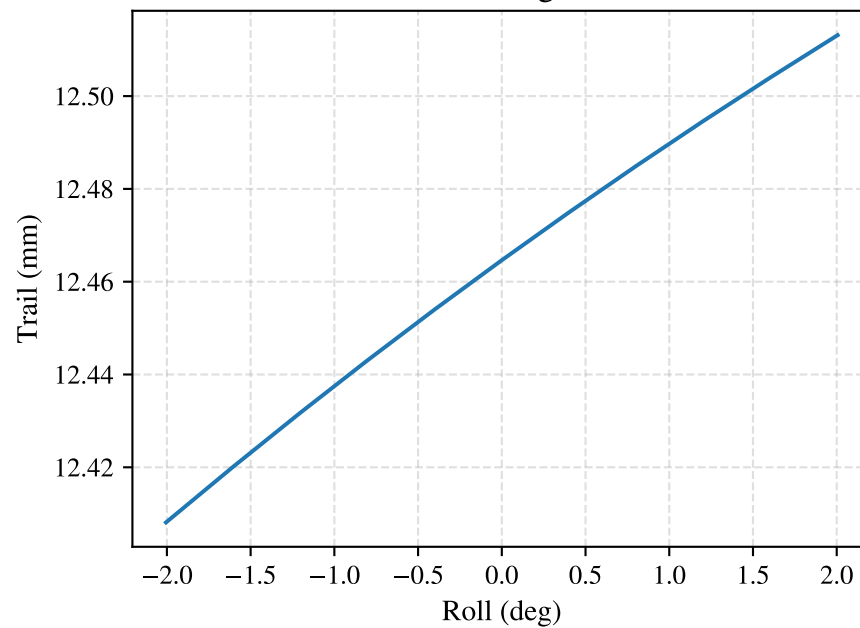


Mechanical Trail vs Roll

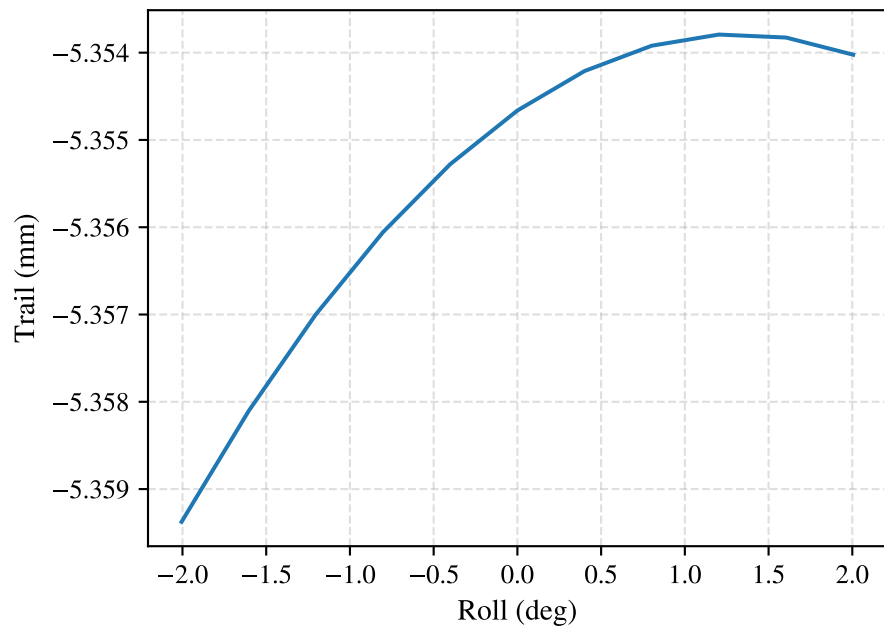
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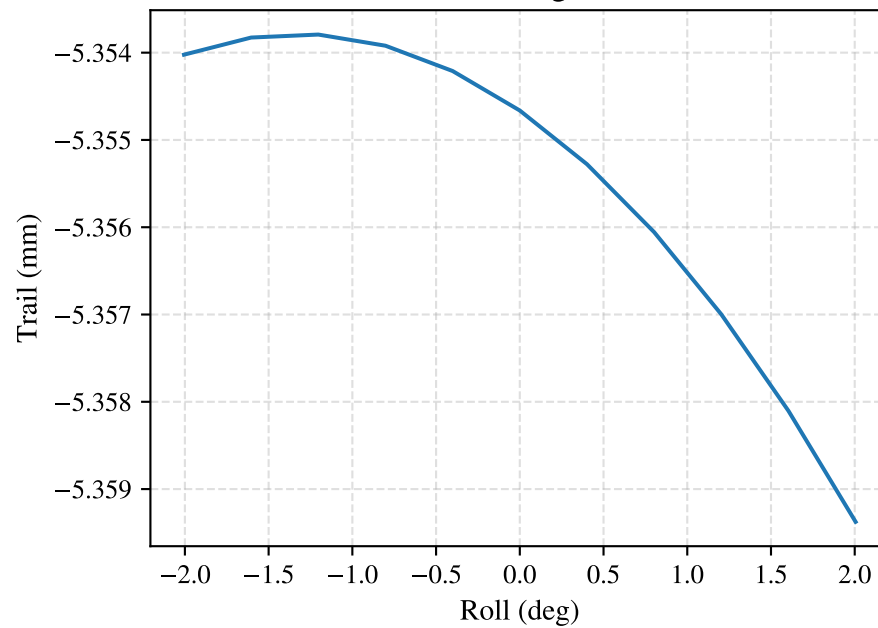
Front Right



Rear Left

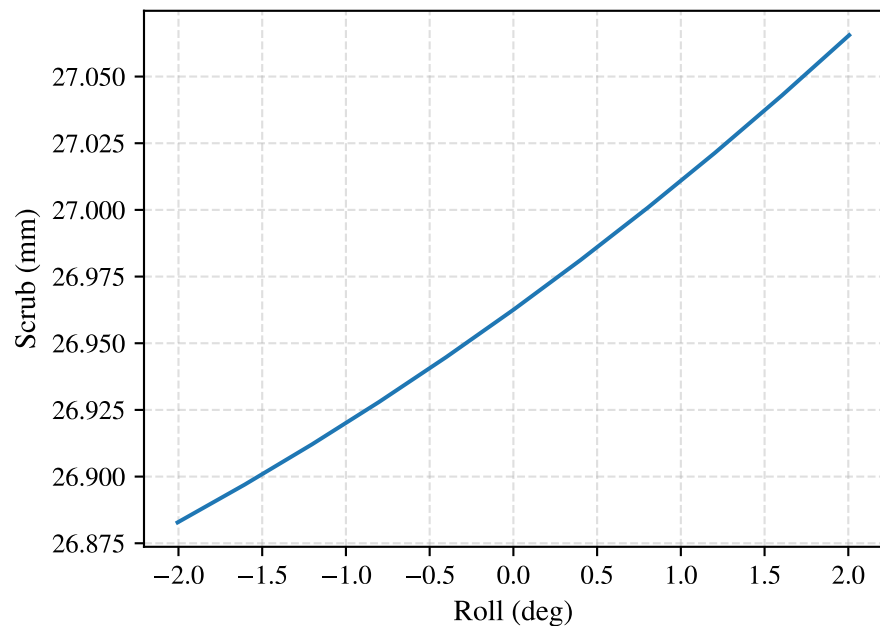


Rear Right

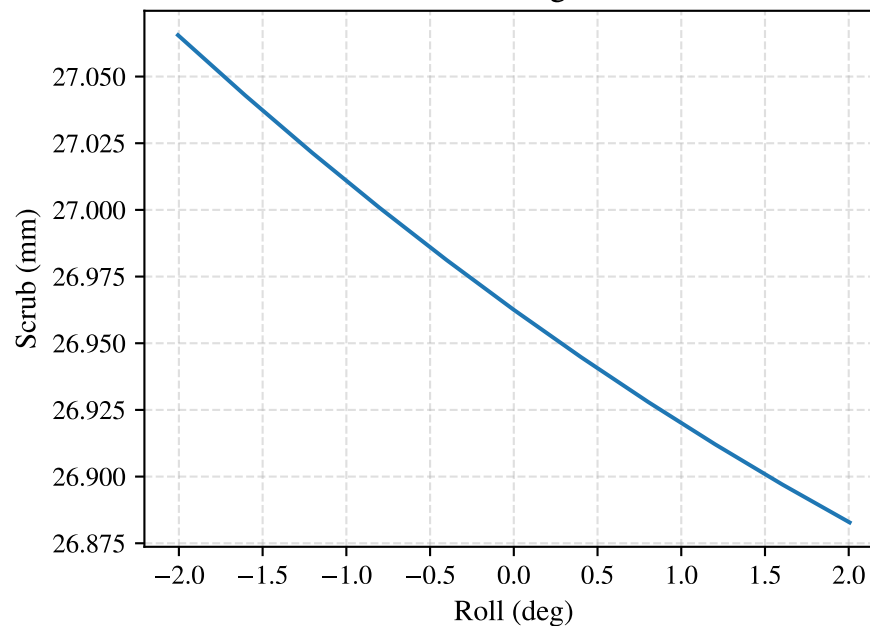


Mechanical Scrub vs Roll

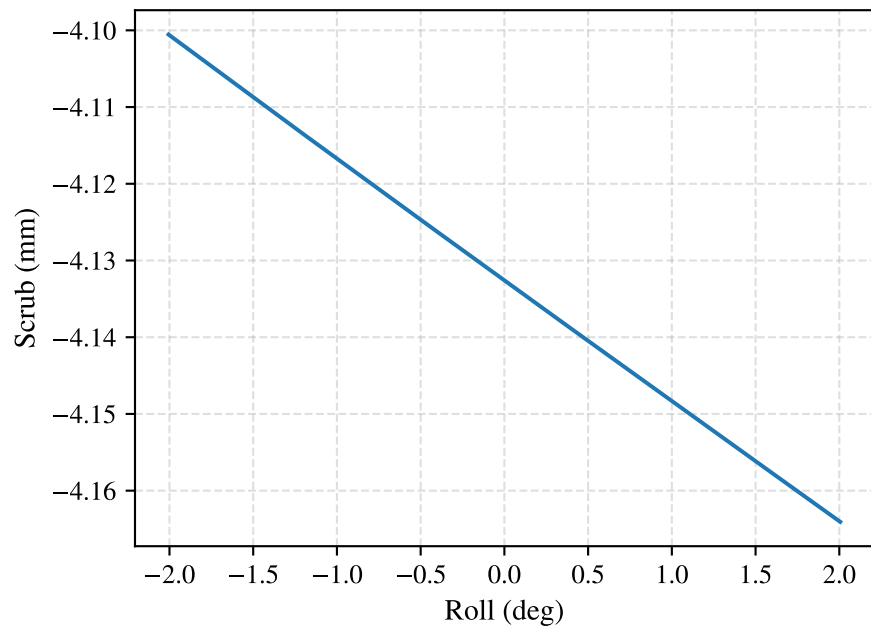
Front Left



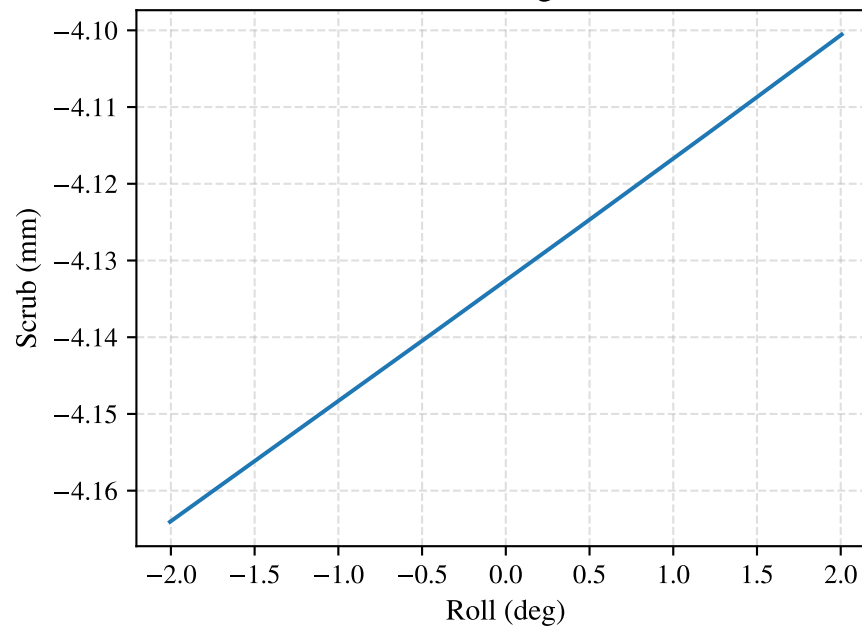
Front Right



Rear Left

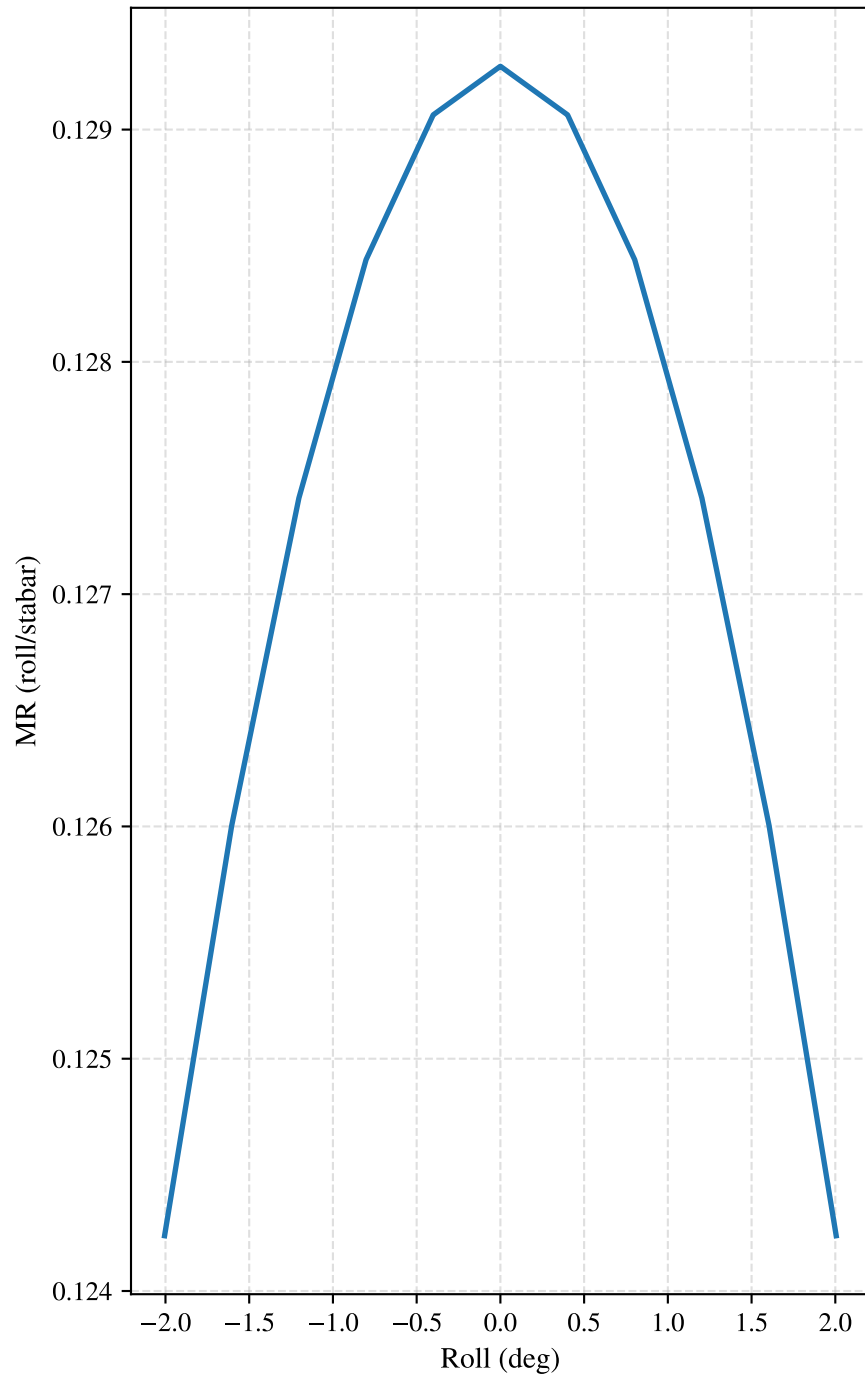


Rear Right

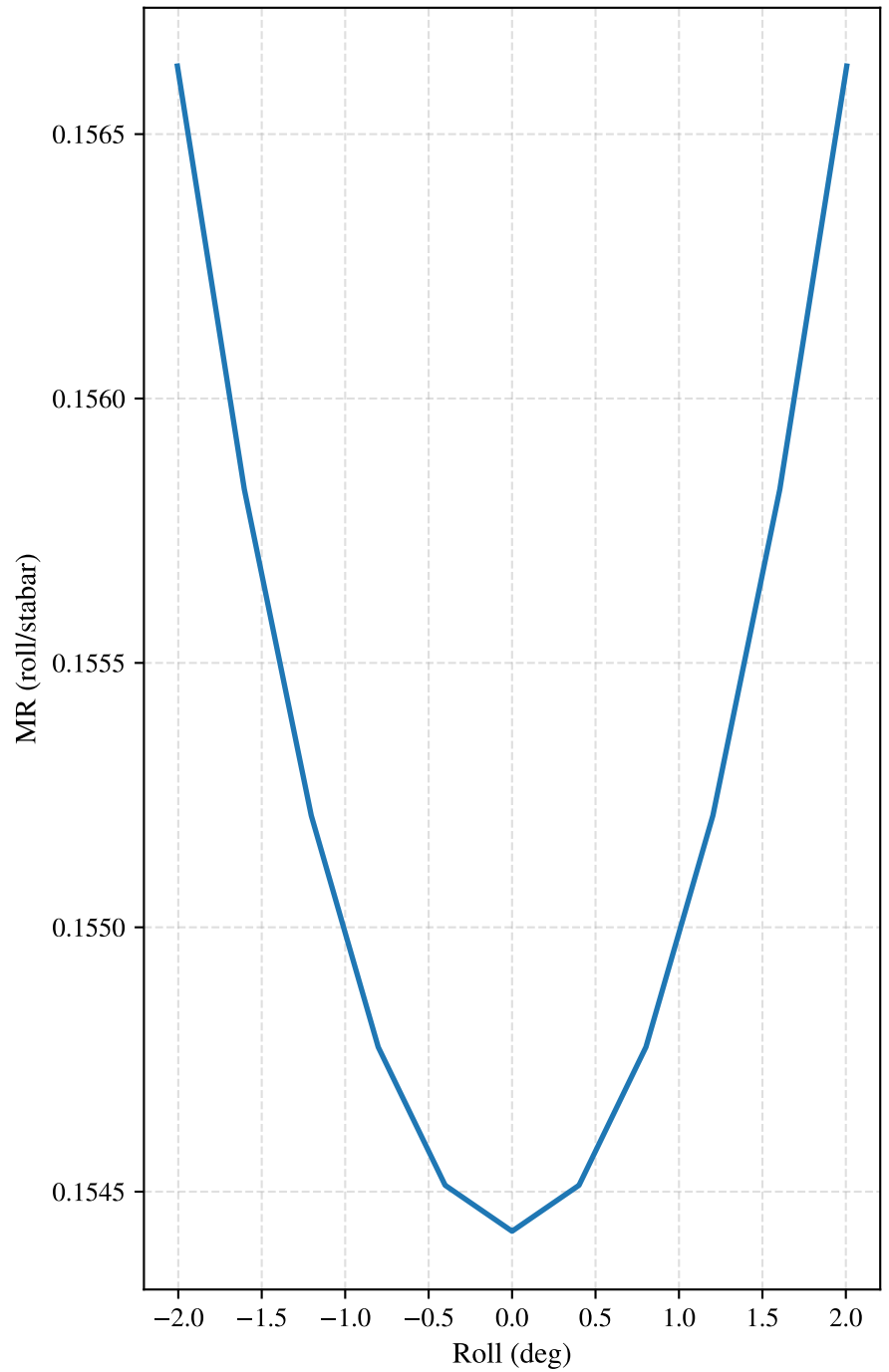


Stabar Motion Ratio vs Roll

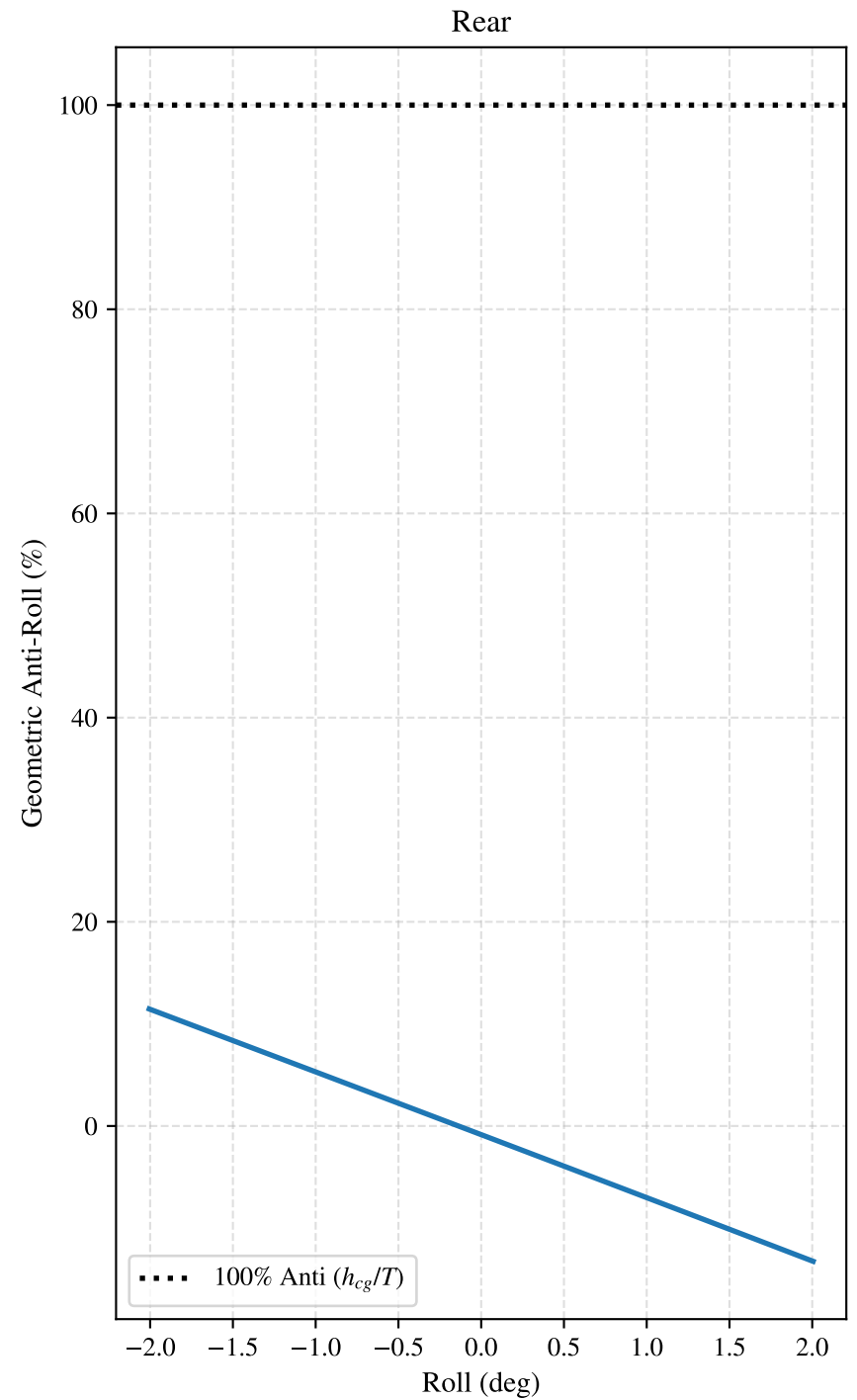
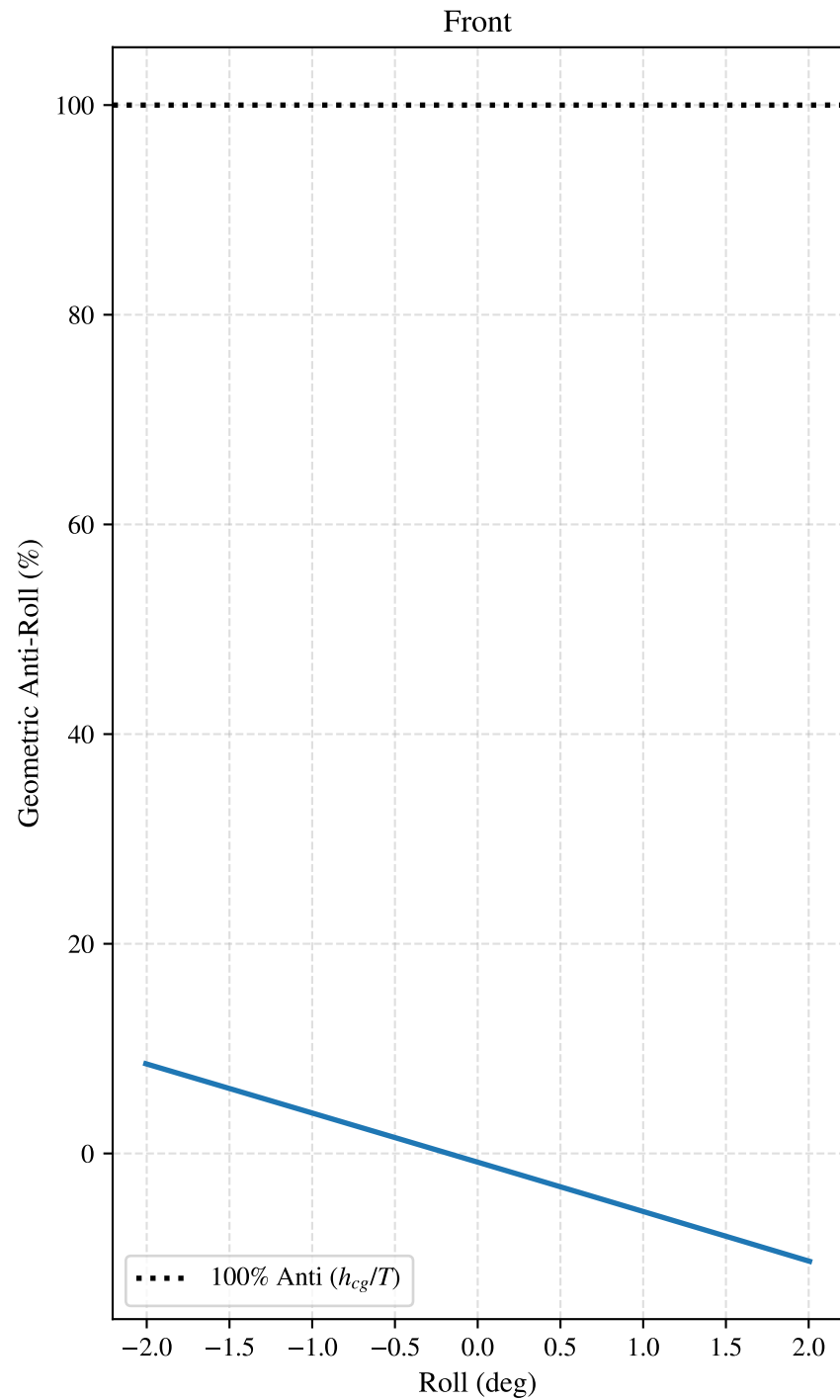
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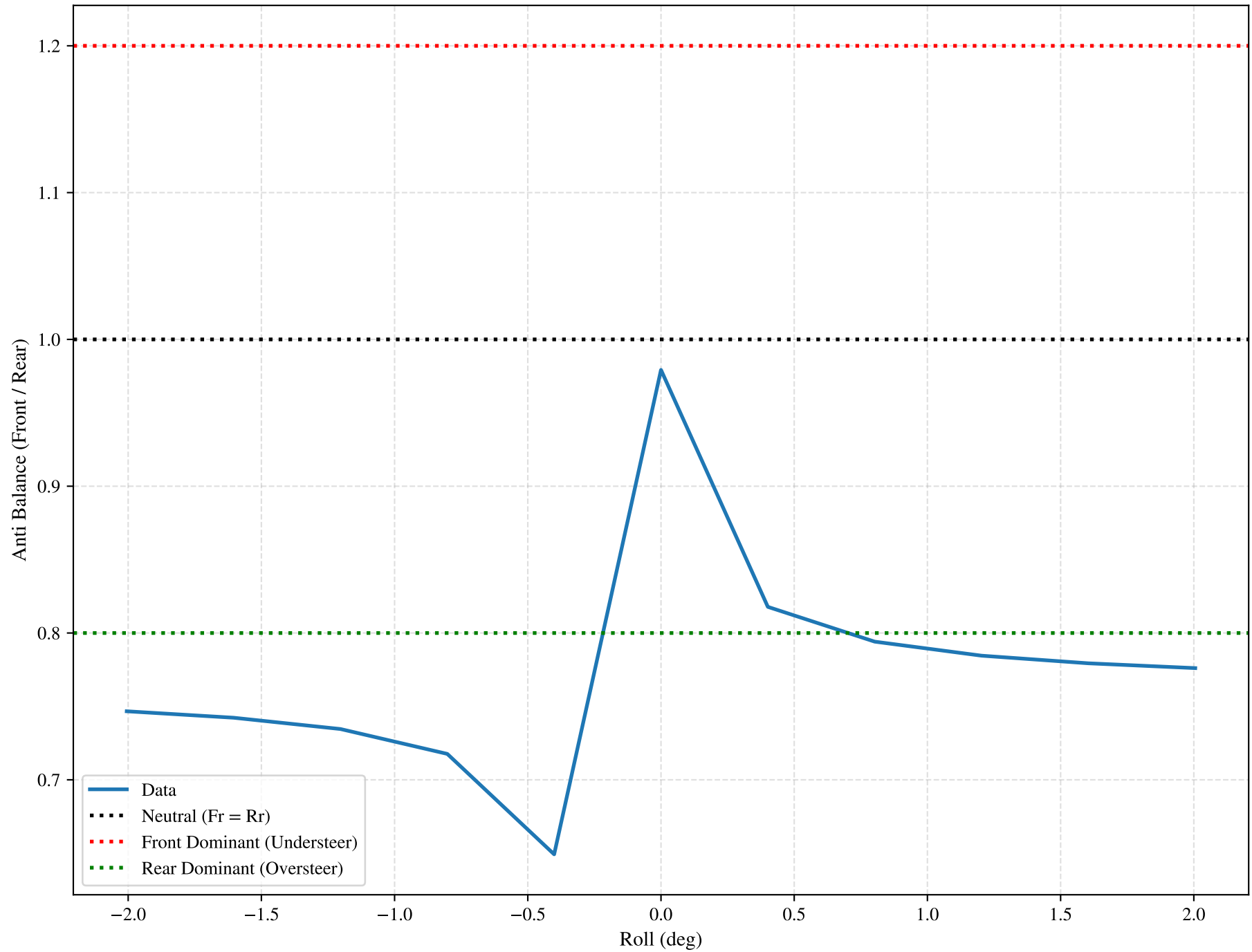
Rear



Geometric Anti-Roll vs Roll



Anti Balance vs Roll



LLTD vs Roll

